

Notes and Exercises for Some Category Theory

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Abstract

This note aims to develop a working knowledge of basic category theory for the study of algebraic geometry and algebraic topology. Set-theoretic issues are ignored. Familiarity with basic categorical concepts, such as functors and natural transformations, as well as with groups, commutative rings, modules, and topological spaces, is assumed. Examples will be given as many as the author can possibly do. The main reference is [4], in which most exercises in Chapter 1 (except the last section) will be worked out here.

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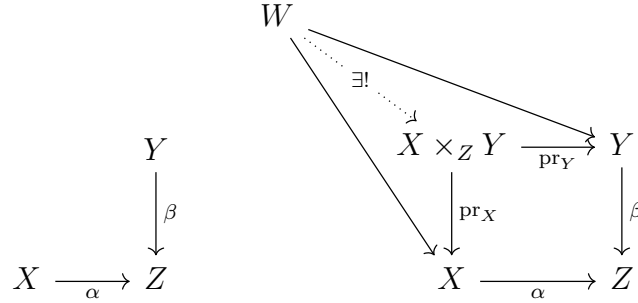
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1 Some Basics

1.1 Pullbacks and Pushouts

Definition 1.1.1. Fix a category $\mathbf{C}$. Given morphisms $\alpha : X \longrightarrow Z$ and $\beta : Y \longrightarrow Z$. The **pull back**, or **fibred product** is an object $X \times_Z Y$, together with morphisms pr_X and pr_Y such that the square diagram commutes. Moreover, this object is universal. That is, for any other object W with morphisms that also satisfy the property, these maps factor through some unique $W \longrightarrow X \times_Z Y$.



The diagram

$$\begin{array}{ccc} X \times_Z Y & \xrightarrow{\text{pr}_Y} & Y \\ \text{pr}_X \downarrow & & \downarrow \beta \\ X & \xrightarrow{\alpha} & Z \end{array}$$

is called a **pullback / fibred / Cartesian diagram / square**.

Example 1.1.2. In **Set**:

$$X \times_Z Y = \{(x, y) \in X \times Y : \alpha(x) = \beta(y)\}$$

with the projections $\text{pr}_X : (x, y) \mapsto x$ and $\text{pr}_Y : (x, y) \mapsto y$.

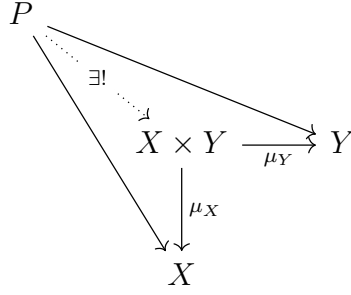
Example 1.1.3. Let X be a topological space. We make a category $\mathcal{O}(X)$ where the objects are the open sets of X and the morphisms are the inclusions (embeddings). That is: $U \longrightarrow V \iff U \subseteq V$. Then:

$$U \times_{\bullet} V = U \cap V$$

with the projections $\text{pr}_U = \text{id}_U$ and $\text{pr}_V = \text{id}_V$.

Example 1.1.4. Let Z be the final object in $\mathbf{C}$, then $X \times_Z Y = X \times Y$. Assume all relevant (fibred) products exist. Here by "=" we mean a unique isomorphism, so they are essentially the same object.

Proof. Recall the universal property of the product $X \times Y$, together with μ_X and μ_Y :



For any other object P with morphisms to X and Y , these morphisms factor through some $P \rightarrow X \times Y$ uniquely. Now, let $P = X \times_Z Y$, with pr_X and pr_Y . By the above universal property of $X \times Y$, $\exists!$ morphism, say $\psi : X \times_Z Y \rightarrow X \times Y$ such that the diagram (left) commutes. In particular, we have:

$$\text{pr}_X = \mu_X \circ \psi$$

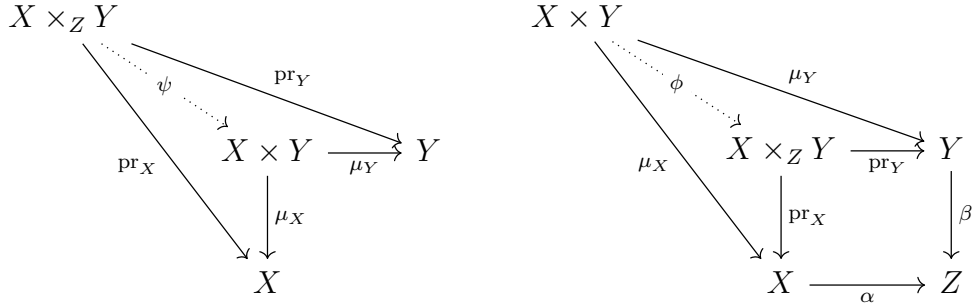
On the other hand, since Z is the final object, $X \times Y$ has a unique morphism to Z , hence $\alpha \circ \mu_X = \beta \circ \mu_Y$. Therefore, by the universal property of $X \times_Z Y$, $\exists!$ morphism, say ϕ , from $X \times Y$ to $X \times_Z Y$ such that the diagram (right) commutes. In particular, we have:

$$\mu_X = \text{pr}_X \circ \phi$$

Therefore, we obtain:

$$\text{pr}_X = \text{pr}_X \circ \phi \circ \psi \quad \text{and} \quad \mu_X = \mu_X \circ \psi \circ \phi$$

Note that $\psi = \text{id}$ and $\phi = \text{id}$ is clearly a solution for this system. By the uniqueness of ψ and ϕ , they have to be the identities, so we are done.



□

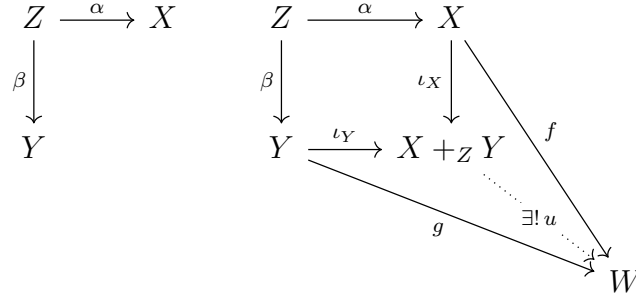
Example 1.1.5 (Magic Diagram). Given morphisms $X_1, X_2 \rightarrow Y$ and $Y \rightarrow Z$. Assume all relevant (fibred) products exist. Show that the following diagram is a Cartesian diagram:

$$\begin{array}{ccc} X_1 \times_Y X_2 & \longrightarrow & X_1 \times_Z X_2 \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y \times_Z Y \end{array}$$

Proof. TBD

□

Definition 1.1.6. Fix a category $\mathbf{C}$. Given morphisms $\alpha : Z \rightarrow X$ and $\beta : Z \rightarrow Y$. The **pushout**, or **fibred coproduct** / **fibred sum**, is an object $X +_Z Y$, together with morphisms ι_X and ι_Y such that the square diagram commutes. Moreover, this object is universal. That is, for any other object W with morphisms $f : X \rightarrow W$ and $g : Y \rightarrow W$ satisfying $f \circ \alpha = g \circ \beta$, these maps factor through some unique $u : X +_Z Y \rightarrow W$.



Example 1.1.7 (This is an interpretation of tensor product of algebras (see [1] page 30)). In **ComRing**, let $A \rightarrow B$ and $A \rightarrow C$ be morphisms. Then B and C can be viewed as A -algebras (so as A -modules). The additive group of $B \otimes_A C$ is the usual tensor-product abelian group, and multiplication is defined on pure tensors by

$$(b_1 \otimes c_1) \cdot (b_2 \otimes c_2) := (b_1 b_2) \otimes (c_1 c_2)$$

with multiplicative identity $1_{B \otimes_A C} = 1_B \otimes 1_C$, then extended to all of $B \otimes_A C$ by bilinearity. Note that we have a natural morphism $\iota_B : B \rightarrow B \otimes_A C$ given by $b \mapsto b \otimes 1_C$, and similar for $\iota_C : C \rightarrow B \otimes_A C$. We claim that $B \otimes_A C$, together with ι_B and ι_C , is the pushout $B +_A C$.

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & B \\ \beta \downarrow & & \downarrow \iota_B \\ C & \xrightarrow{\iota_C} & B \otimes_A C \end{array}$$

Proof. First, the square commutes: For all $a \in A$:

$$(\iota_B \circ \alpha)(a) = \alpha(a) \otimes 1_C = a \bullet (1_B \otimes 1_C) = 1_B \otimes \beta(a) = (\iota_C \circ \beta)(a)$$

Now let $R \in \text{Obj}(\mathbf{ComRing})$ and ring morphisms $f : B \rightarrow R$, $g : C \rightarrow R$ satisfy

$$f \circ \alpha = g \circ \beta$$

Define

$$\begin{aligned} \tilde{h} : B \times C &\longrightarrow R \\ (b, c) &\longmapsto f(b)g(c) \end{aligned}$$

Convince yourself that this map is A -bilinear. Hence, by the universal property of tensor products, there exists a unique A -module homomorphism

$$h : B \otimes_A C \longrightarrow R$$

such that $h(b \otimes c) = f(b)g(c)$.

We claim h is also a ring morphism. Indeed, on pure tensors,

$$\begin{aligned}
& h((b_1 \otimes c_1)(b_2 \otimes c_2)) \\
&= h((b_1 b_2) \otimes (c_1 c_2)) \\
&= f(b_1 b_2)g(c_1 c_2) \\
&= f(b_1)g(c_1)f(b_2)g(c_2) \\
&= h(b_1 \otimes c_1) h(b_2 \otimes c_2)
\end{aligned}$$

and $h(1_B \otimes 1_C) = f(1_B)g(1_C) = 1_R$, as desired.

Also

$$h \circ \iota_B = f \quad h \circ \iota_C = g$$

since $h(\iota_B(b)) = h(b \otimes 1_C) = f(b)$ and similarly for C .

Therefore, we obtain the following diagram in **ComRing**:

$$\begin{array}{ccc}
A & \xrightarrow{\alpha} & B \\
\downarrow \beta & & \downarrow \iota_B \\
C & \xrightarrow{\iota_C} & B \otimes_A C \\
& \searrow g & \nearrow f \\
& & R
\end{array}$$

(Note: A dotted arrow labeled h points from $B \otimes_A C$ to R .)

Finally, uniqueness: if $h' : B \otimes_A C \rightarrow R$ is another ring morphism with $h' \circ \iota_B = f$ and $h' \circ \iota_C = g$, then for any pure tensor,

$$h'(b \otimes c) = h'((b \otimes 1_C)(1_B \otimes c)) = h'(b \otimes 1_C) h'(1_B \otimes c) = f(b)g(c) = h(b \otimes c)$$

Hence $h' = h$.

Therefore, $B \otimes_A C$, together with ι_B and ι_C , satisfies the universal property of the pushout. So

$$B \otimes_A C \cong B +_A C$$

□

1.2 Limits and Colimits

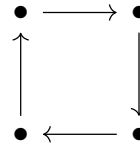
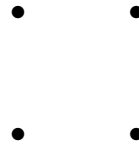
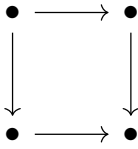
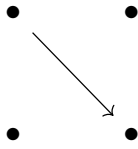
Definition 1.2.1. Let $\mathbf{I}$ be a small category, $\mathbf{C}$ be an arbitrary category. Then a functor $F : \mathbf{I} \longrightarrow \mathbf{C}$ is said to be a **diagram** indexed by $\mathbf{I}$. We call $\mathbf{I}$ an **index category**. Our index categories will usually be partially ordered sets, so that there is at most one morphism between two different objects in $\mathbf{I}$. In that sense, the name [index] has a concrete meaning, and we may draw the diagram explicitly. The following are all valid examples for $\mathbf{I}$: (We may ignore drawing all the compositions of morphisms and all the identities).

$$\cdots \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \bullet$$

$$\bullet \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \cdots$$

$$\cdots \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \cdots$$

If we restrict to poset-indexed diagrams, the first three of the following are still valid examples for $\mathbf{I}$, but the last one is not:



Then the **limit of the diagram** is an object $\varprojlim_{\mathbf{I}} A_i$, together with morphisms $f_j : \varprojlim_{\mathbf{I}} A_i \longrightarrow A_j$ for each $j \in \text{Obj}(\mathbf{I})$, such that if $m \in \text{hom}_{\mathbf{I}}(j, k)$, then the following diagram commutes:

$$\begin{array}{ccc} & \varprojlim_{\mathbf{I}} A_i & \\ f_j \swarrow & & \searrow f_k \\ A_j & \xrightarrow{F(m)} & A_k \end{array}$$

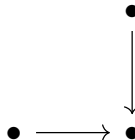
And this object $\varprojlim_{\mathbf{I}} A_i$, together with $\{f_j\}$, are universal w.r.t. this property. Namely, given any other object $\Omega \in \text{Obj}(\mathbf{C})$, together with $\{g_i\}$, that also satisfy this property, then $\exists!$ morphism $\phi : \Omega \longrightarrow \varprojlim_{\mathbf{I}} A_i$. And hence we can show that the limit $\varprojlim_{\mathbf{I}} A_i$, if it exists, is unique up to unique isomorphism.

Example 1.2.2. If the diagram $\mathbf{I}$ is:



Then we obtain the product. More generally, if $\mathbf{I}$ is a set, i.e., the only morphisms are identities, then $\varprojlim_{\mathbf{I}} A_i = \prod A_i$.

Example 1.2.3. If the diagram $\mathbf{I}$ is:



Then we obtain the fibred product.

Example 1.2.4. If $\mathbf{I}$ has an initial object e , then $\varprojlim_{\mathbf{I}} A_i = A_e$.

Example 1.2.5. In **Set**,

$$\left\{ (a_i)_{i \in \mathbf{I}} \in \prod_i A_i : F(m)(a_j) = a_k, \forall m \in \text{hom}_{\mathbf{I}}(j, k) \right\}$$

along with the obvious projection maps to each A_i , is the limit $\varprojlim_{\mathbf{I}} A_i$. That is, each element in the limit can be viewed as a sequence (or a family, in general) of elements of each object indexed by $\mathbf{I}$, that are compatible with the morphisms. This interpretation is available to all the categories where the objects are set-like.

The following examples have index category $\mathbf{I} : \cdots \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \bullet$.

Example 1.2.6 (p -adic integers). Fix a prime number p . The p -**adic integers**, denoted as $\mathbb{Z}_p$, are described informally as $\{a_0 + a_1p + a_2p^2 + \cdots \mid a_i \in \mathbb{Z} \text{ and } 0 \leq a_i \leq p\}$. In **ComRing**:

$$\begin{array}{c} \mathbb{Z}_p = \varprojlim (\mathbb{Z}/p^i\mathbb{Z}) \\ \swarrow \quad \searrow \quad \searrow \quad \searrow \\ \cdots \longrightarrow \mathbb{Z}/p^3\mathbb{Z} \longrightarrow \mathbb{Z}/p^2\mathbb{Z} \longrightarrow \mathbb{Z}/p\mathbb{Z} \end{array}$$

where

$$\begin{array}{c} a_0 + a_1p + a_2p^2 + a_3p^3 + \cdots \\ \swarrow \quad \searrow \quad \searrow \quad \searrow \\ \cdots \longrightarrow a_0 + a_1p + a_2p^2 \longrightarrow a_0 + a_1p \longrightarrow a_0 \end{array}$$

inspired by the previous example, the element $a_0 + a_1p + a_2p^2 + a_3p^3 + \cdots \in \mathbb{Z}_p$ can be viewed as the sequence $\{a_0, a_0 + a_1p, a_0 + a_1p + a_2p^2, \dots\}$.

The following three examples are from [2].

Example 1.2.7 (Formal power series). Similarly, recall the ring of **formal power series** $R[[x]] = \{r_0 + r_1x + r_2x^2 + \cdots \mid r_i \in R\}$. In **ComRing**, it is the limit

$$\begin{array}{c} R[[x]] \\ \swarrow \quad \searrow \quad \searrow \quad \searrow \\ \cdots \longrightarrow R[x] / \langle x^3 \rangle \longrightarrow R[x] / \langle x^2 \rangle \longrightarrow R[x] / \langle x \rangle \end{array}$$

Example 1.2.8 (Infimum). Consider $\mathbf{I} \longrightarrow (\mathbb{R}, \leq)$. The limit, if it exists, is exactly the infimum. For example:

$$\begin{array}{c}
 0 = \inf_{k \geq 1} \left\{ \frac{1}{2^k} \right\} \\
 \swarrow \quad \searrow \quad \searrow \\
 \dots \longrightarrow 1/8 \longrightarrow 1/4 \longrightarrow 1/2
 \end{array}$$

Example 1.2.9 (Symmetric polynomials). Assume the notion of graded rings. The ring of symmetric functions Λ can be defined as the limit (in the category of graded rings) of the inverse system of symmetric polynomial rings where p_i is induced by setting $x_i \mapsto 0$.

$$\begin{array}{c}
 \Lambda \\
 \swarrow \quad \searrow \quad \searrow \\
 \dots \longrightarrow \mathbb{Z}[x_1, x_2, x_3]^{S_3} \xrightarrow{p_3} \mathbb{Z}[x_1, x_2]^{S_2} \xrightarrow{p_2} \mathbb{Z}[x_1]^{S_1}
 \end{array}$$

and Λ is graded by total degree:

$$\Lambda = \bigoplus_{d \geq 0} \Lambda^d.$$

Definition 1.2.10. Dually, given the index category $\mathbf{I}$, the **colimit of the diagram** is an object $\varinjlim_{\mathbf{I}} A_i$, together with morphisms $h_j : A_j \longrightarrow \varinjlim_{\mathbf{I}} A_i$ for each $j \in \text{Obj}(\mathbf{I})$, such that if $m \in \text{hom}_{\mathbf{I}}(j, k)$, then the following diagram commutes:

$$\begin{array}{ccc}
 A_j & \xrightarrow{F(m)} & A_k \\
 & \searrow h_j \quad \swarrow h_k & \\
 & \varinjlim_{\mathbf{I}} A_i &
 \end{array}$$

And this object $\varinjlim_{\mathbf{I}} A_i$, together with $\{h_j\}$, are universal w.r.t. this property. Namely, given any other object $\Omega \in \text{Obj}(\mathbf{C})$, together with $\{g_i\}$, that also satisfy this property, then $\exists!$ morphism $\psi : \varinjlim_{\mathbf{I}} A_i \longrightarrow \Omega$ such that $\psi \circ h_i = g_i$ for all $i \in \text{Obj}(\mathbf{I})$.

$$\begin{array}{ccc}
 A_i & \xrightarrow{h_i} & \varinjlim_{\mathbf{I}} A_i \\
 & \searrow g_i & \downarrow \exists! \psi \\
 & & \Omega
 \end{array}$$

And hence we can show that the colimit $\varinjlim_{\mathbf{I}} A_i$, if it exists, is unique up to unique isomorphism.

Example 1.2.11. Let k be a fixed positive integer, then the set $k^{-\infty}\mathbb{Z}$ (actually a fractional ring) of rational numbers whose denominators are powers of k is a colimit $\varinjlim k^{-i}\mathbb{Z}$, provided the diagram:

$$\mathbb{Z} \longrightarrow k^{-1}\mathbb{Z} \longrightarrow k^{-2}\mathbb{Z} \longrightarrow \dots$$

Indeed, we have:

$$\begin{array}{ccc}
 k^{-m}\mathbb{Z} & \xrightarrow{\iota_{m,n}} & k^{-n}\mathbb{Z} \\
 & \searrow \iota_m \quad \swarrow \iota_n & \\
 & k^{-\infty}\mathbb{Z} = \varinjlim k^{-i}\mathbb{Z} &
 \end{array}$$

where all the morphisms are just embeddings. $\iota_{m,n}$ is defined if and only if $m \leq n$.

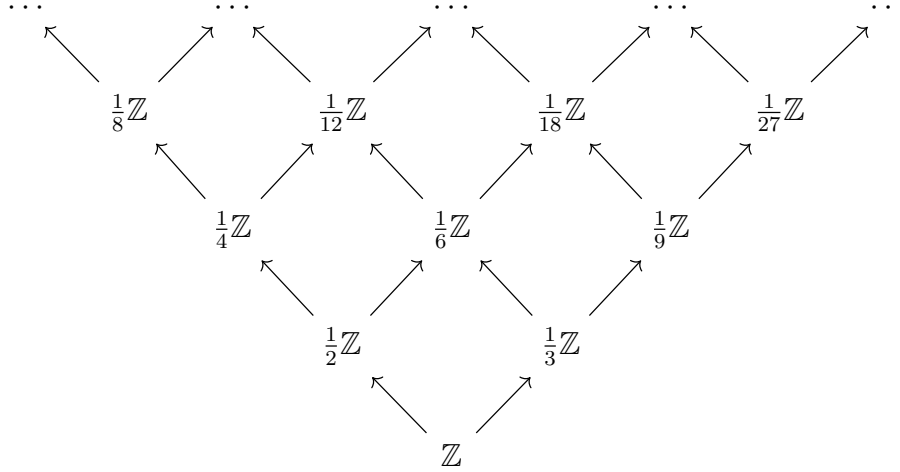
Example 1.2.12. The previous example can be generalized to:

$$\mathbb{Q} = \varinjlim_n \frac{1}{n}\mathbb{Z}$$

Here the index category is the poset $(\mathbb{Z}_{>0}, \preceq)$: for $m, n \in \mathbb{Z}_{>0}$, we set $m \preceq n$ iff $m \mid n$, and the transition morphisms are the natural embeddings

$$\frac{1}{m}\mathbb{Z} \hookrightarrow \frac{1}{n}\mathbb{Z}$$

For the convenience of drawing a diagram, we only display the part generated by two prime integers, say 2 and 3:



Again, each morphism is just an embedding. The displayed 2-3 sub-diagram has colimit $\mathbb{Z}[1/6]$. After extending the diagram to all positive integers (equivalently, all prime directions), the colimit is exactly $\mathbb{Q}$.

Example 1.2.13. Let X be a fixed set, and let $\{A_i\}_{i \in I}$ be a family of subsets of X . Consider $\mathcal{P}(X)$ as a category whose objects are subsets of X , and whose morphisms are inclusions. Regard $\{A_i\}$ as a diagram in $\mathcal{P}(X)$ (with the obvious inclusion maps among them). Then their union

$$\bigcup_{i \in I} A_i$$

is the colimit of this diagram in $\mathcal{P}(X)$ (equivalently in **Set**): each A_i maps to $\bigcup_i A_i$ by inclusion, and $\bigcup_i A_i$ is the smallest subset of X receiving all these maps.

Colimits do not always exist, but there are two large classes of examples for which they do.

Definition 1.2.14. A non-empty poset $(S, \succeq)$ is **filtered** iff for each $x, y \in S$, $\exists z$ s.t. $x \succeq z$ and $y \succeq z$. More generally, a nonempty category $\mathbf{J}$ is **filtered** iff:

- (i) For each $x, y \in \text{Obj}(\mathbf{J})$, $\exists z \in \text{Obj}(\mathbf{J})$ and morphisms $x \longrightarrow z$ and $y \longrightarrow z$. Essentially, this means that every two objects have a common "upper bound".
- (ii) For every two morphisms $u, v : x \longrightarrow y$, $\exists w : y \longrightarrow z$ such that $w \circ u = w \circ v$.

Example 1.2.15. Here are some examples of categories that are not filtered:

- (1) Discrete categories. (No morphisms except the identities.)
- (2) A category with objects x, y and two distinct parallel arrows $u, v : x \longrightarrow y$, and no non-identity arrows out of y . This fails **(ii)**: there is no $w : y \longrightarrow z$ such that $w \circ u = w \circ v$.
- (2) A poset as a category where x, y are maximal and incomparable. This fails **(i)**.
- (4) $\mathbf{BG}$ where G is non-trivial. (See the definition in example 2.4.3.) Condition **(ii)** would require: for all u, v , there exists w with $wu = wv$, which forces $u = v$ by left cancellation. Hence, only the trivial group gives a filtered category.

Theorem 1.2.16. Let $F : \mathbf{J} \longrightarrow \mathbf{Set}$ be a diagram indexed by a filtered category $\mathbf{J}$, and write $A_i := F(i)$ for each $i \in \text{Obj}(\mathbf{J})$. Then its colimit can be interpreted as

$$\varinjlim_{\mathbf{J}} A_i = \left(\bigsqcup_{i \in \text{Obj}(\mathbf{J})} A_i \right) / \sim$$

where $\bigsqcup$ is the disjoint union, and the relation $\sim$ is defined by

$$(a_i, i) \sim (a_j, j) \iff \exists k \in \text{Obj}(\mathbf{J}), \exists f : i \longrightarrow k, g : j \longrightarrow k, \text{ s.t. } F(f)(a_i) = F(g)(a_j) \in A_k.$$

The canonical maps $\iota_i : A_i \longrightarrow \varinjlim_{\mathbf{J}} A_i$ send a_i to the class $[(a_i, i)]$.

Proof. First, we show that $\sim$ is an equivalence relation. Reflexivity and symmetry are immediate. We prove transitivity. Suppose

$$(a_i, i) \sim (a_j, j) \quad \text{and} \quad (a_j, j) \sim (a_k, k)$$

Then there exist objects ℓ, m in $\mathbf{J}$ and morphisms

$$\mu_{i\ell} : i \longrightarrow \ell \quad \mu_{j\ell} : j \longrightarrow \ell \quad \mu_{jm} : j \longrightarrow m \quad \mu_{km} : k \longrightarrow m$$

such that

$$F(\mu_{i\ell})(a_i) = F(\mu_{j\ell})(a_j) \in A_\ell \quad F(\mu_{jm})(a_j) = F(\mu_{km})(a_k) \in A_m$$

By filtered condition **(i)**, there exists $n \in \text{Obj}(\mathbf{J})$ with arrows

$$\mu_{\ell n} : \ell \longrightarrow n \quad \mu_{mn} : m \longrightarrow n$$

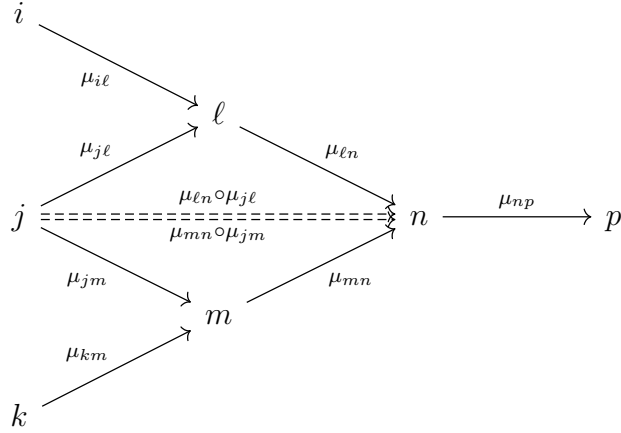
Hence

$$F(\mu_{\ell n} \circ \mu_{i\ell})(a_i) = F(\mu_{\ell n} \circ \mu_{j\ell})(a_j) \quad F(\mu_{mn} \circ \mu_{jm})(a_j) = F(\mu_{mn} \circ \mu_{km})(a_k)$$

Now $\mu_{\ell n} \circ \mu_{j\ell}, \mu_{mn} \circ \mu_{jm} : j \longrightarrow n$ are parallel arrows. By filtered condition **(ii)**, there exist $p \in \text{Obj}(\mathbf{J})$ and $\mu_{np} : n \longrightarrow p$ such that

$$\mu_{np} \circ \mu_{\ell n} \circ \mu_{j\ell} = \mu_{np} \circ \mu_{mn} \circ \mu_{jm}$$

Diagrammatically:



Applying F and composing equalities, we get

$$F(\mu_{np} \circ \mu_{\ell n} \circ \mu_{i\ell})(a_i) = F(\mu_{np} \circ \mu_{\ell n} \circ \mu_{j\ell})(a_j) = F(\mu_{np} \circ \mu_{mn} \circ \mu_{jm})(a_j) = F(\mu_{np} \circ \mu_{mn} \circ \mu_{km})(a_k)$$

Therefore $(a_i, i) \sim (a_k, k)$, so $\sim$ is transitive.

Now, denote $\left(\bigsqcup_{i \in \text{Obj}(\mathbf{J})} A_i\right) / \sim$ as $\mathbf{A}$. We show that $\mathbf{A}$ is indeed the colimit $\varinjlim_{\mathbf{J}} A_i$:

Take any $i, j \in \text{Obj}(\mathbf{J})$, together with an arbitrary morphism $\mu_{ij} : i \longrightarrow j$. First, we need to show that the following diagram commutes:

$$\begin{array}{ccc}
 A_i & \xrightarrow{F(\mu_{ij})} & A_j \\
 \searrow \iota_i & & \swarrow \iota_j \\
 & \mathbf{A} &
 \end{array}$$

That is, for any $a_i \in A_i$, we need to show:

$$(a_i, i) \sim (F(\mu_{ij})(a_i), j)$$

Indeed, we may take $f = \mu_{ij}$ and $g = \text{id}_j$. Let $a_j := F(\mu_{ij})(a_i)$. Then

$$F(f)(a_i) = F(\mu_{ij})(a_i) = F(\text{id}_j)(a_j) = F(g)(a_j)$$

as desired. Now let $\mathbf{B} \in \text{Obj}(\mathbf{Set})$ and assume we are given a compatible family

$$\{\theta_i : A_i \longrightarrow \mathbf{B}\}_{i \in \text{Obj}(\mathbf{J})} \quad \theta_j \circ F(\mu_{ij}) = \theta_i \text{ for all } \mu_{ij} : i \longrightarrow j$$

Define

$$h : \mathbf{A} \longrightarrow \mathbf{B}, \quad h([(a_i, i)]) := \theta_i(a_i)$$

We check well-definedness: if $(a_i, i) \sim (a_j, j)$, then there exist $k \in \text{Obj}(\mathbf{J})$ and arrows $f : i \longrightarrow k$, $g : j \longrightarrow k$ such that $F(f)(a_i) = F(g)(a_j)$. By compatibility,

$$\theta_i(a_i) = \theta_k(F(f)(a_i)) = \theta_k(F(g)(a_j)) = \theta_j(a_j)$$

So h is well-defined. By definition of h , for each i and each $a_i \in A_i$,

$$(h \circ \iota_i)(a_i) = h([(a_i, i)]) = \theta_i(a_i)$$

hence $h \circ \iota_i = \theta_i$. Therefore the diagram

$$\begin{array}{ccc} A_i & \xrightarrow{\iota_i} & \mathbf{A} \\ & \searrow \theta_i & \downarrow \exists h \\ & & \mathbf{B} \end{array}$$

commutes for all i .

Finally, uniqueness: if $h' : \mathbf{A} \longrightarrow \mathbf{B}$ also satisfies $h' \circ \iota_i = \theta_i$ for all i , then for any class $[(a_i, i)]$,

$$h'([(a_i, i)]) = h'(\iota_i(a_i)) = \theta_i(a_i) = h([(a_i, i)]).$$

Hence $h' = h$. So $\mathbf{A}$ satisfies the universal property of $\varinjlim_{\mathbf{J}} A_i$ □

Example 1.2.17 (Colimit of A -modules). The colimit $\varinjlim M_i$ in the category $\mathbf{Mod}_A$ can be described as follows: The underlying set is defined as in Theorem 1.2.16. Let $m_i \in M_i, m_j \in M_j$, since our index category $\mathbf{J}$ is filtered, $\exists k$ with $\theta_{ik} : M_i \longrightarrow M_k$ and $\theta_{jk} : M_j \longrightarrow M_k$. Then:

$$[(m_i, i)] + [(m_j, j)] := [(\theta_{ik}(m_i) + \theta_{jk}(m_j), k)]$$

And the A -linear multiplication is defined in the obvious way.

Example 1.2.18 (Localization as a colimit). This is the generalization of Example 1.2.12. Suppose that A is an integral domain. Find a filtered index category $\mathbf{J}$ such that in $\mathbf{Mod}_A$, we have

$$S^{-1}A = \varinjlim_S \frac{1}{s}A$$

Just as in the previous example of integers, we may define the index category to be the poset $(S, \preceq)$: for $s_1, s_2 \in S$, we set $s_1 \preceq s_2$ iff $s_1 \mid s_2$, i.e. $\exists t \in S$ such that $s_2 = ts_1$, and the transition morphisms are the natural embeddings

$$\frac{1}{s_1}A \hookrightarrow \frac{1}{s_2}A$$

More generally, even if A is not an integral domain, we can still interpret $S^{-1}A$ as a colimit in the category of A -algebras $\mathbf{Alg}_A$, as follows: Let $A_s := A[1/s]$ for each $s \in S$. Then

$$S^{-1}A = \varinjlim A_s$$

Example 1.2.19 (Colimit of A -modules without the filtered condition). Given a diagram of A -modules indexed by $\mathbf{I}$. Let $M_i = F(i)$. We claim that:

$$\varinjlim_{\mathbf{I}} M_i = \left(\bigoplus_{i \in \text{Obj}(\mathbf{I})} M_i \right) / D$$

where D is the submodule of $\bigoplus_{i \in \text{Obj}(\mathbf{I})} M_i$ generated by $m_i - f(u)(m_i)$ for every $u : i \longrightarrow j$.

Proof. See [1] Exercise 16, Chapter 2. □

2 More Basics

Throughout this section, $\mathbf{C}$ and $\mathbf{D}$ will denote locally small categories. $U(-)$ will denote the forgetful functor (The exact meaning depends on the context). R will denote a commutative ring with unity.

2.1 The Yoneda Embedding

Fix two arbitrary objects $A, A' \in \text{Obj}(\mathbf{C})$. Denote the two functors $\text{hom}_{\mathbf{C}}(-, A), \text{hom}_{\mathbf{C}}(-, A')$ as $h_A, h_{A'}$, respectively. Is there any connection between $\text{Nat}(h_A, h_{A'})$ and $\text{hom}_{\mathbf{C}}(A, A')$?

Theorem 2.1.1 (Yoneda Embedding). *Under the above setting, for any natural transformation $i : h_A \Rightarrow h_{A'}$, there exists a unique morphism $g \in \text{hom}_{\mathbf{C}}(A, A')$ (depending on the choice of i , of course) such that for any $C \in \text{Obj}(\mathbf{C})$ and any $u \in \text{hom}_{\mathbf{C}}(C, A)$,*

$$i_C(u) = g \circ u$$

Moreover, if for any $C \in \text{Obj}(\mathbf{C})$ the sets $\text{hom}_{\mathbf{C}}(C, A)$ and $\text{hom}_{\mathbf{C}}(C, A')$ are always naturally isomorphic, then g is an isomorphism.

Proof. Fix an arbitrary natural transformation i . We first show the existence of such a $g \in \text{hom}_{\mathbf{C}}(A, A')$. Let

$$g := i_A(\text{id}_A)$$

Now, pick any $C \in \text{Obj}(\mathbf{C})$. Since i is a natural transformation, the following diagram commutes:

$$\begin{array}{ccc}
 \text{hom}_{\mathbf{C}}(A, A) & \xrightarrow{i_A} & \text{hom}_{\mathbf{C}}(A, A') \\
 \downarrow - \circ u & & \downarrow - \circ u \\
 & \begin{array}{ccc} \text{id}_A & \mapsto & g \\ \downarrow & & \downarrow \\ u & \mapsto & g \circ u \end{array} & \\
 \text{hom}_{\mathbf{C}}(C, A) & \xrightarrow{i_C} & \text{hom}_{\mathbf{C}}(C, A')
 \end{array}$$

This forces i_C to map u to $g \circ u$. To show uniqueness, suppose there exists $g' \in \text{hom}(A, A')$ satisfying the property. Then $i_A(\text{id}_A) = g' \circ \text{id}_A = g'$, but we also have $i_A(\text{id}_A) = g$, hence $g = g'$. Now, suppose $\text{hom}(C, A) \cong \text{hom}(C, A')$ via i_C for all $C \in \text{Obj}(\mathbf{C})$. Consider $A' \in \text{Obj}(\mathbf{C})$; the

following diagram commutes:

$$\begin{array}{ccc}
\mathrm{hom}_{\mathbf{C}}(A', A) & \xrightarrow{g \circ -} & \mathrm{hom}_{\mathbf{C}}(A', A') \\
\downarrow - \circ g & & \downarrow - \circ g \\
& \begin{array}{ccc} f & \xrightarrow{\quad} & \mathrm{id}_{A'} \\ \downarrow & & \downarrow \\ \mathrm{id}_A & \xrightarrow{\quad} & g \end{array} & & \\
\mathrm{hom}_{\mathbf{C}}(A, A) & \xrightarrow{g \circ -} & \mathrm{hom}_{\mathbf{C}}(A, A')
\end{array}$$

Take $\mathrm{id}_{A'} \in \mathrm{hom}_{\mathbf{C}}(A', A')$; since $i_{A'}$ is an isomorphism, there exists an element in $\mathrm{hom}_{\mathbf{C}}(A', A)$, say f , such that $i_{A'}(f) = g \circ f = \mathrm{id}_{A'}$. The diagram forces $f \circ g = \mathrm{id}_A$. Therefore, since we have exhibited a two-sided inverse, g is an isomorphism, as desired. $\square$

Remark. The second part of the theorem says the following: let the object C run through all of $\mathrm{Obj}(\mathbf{C})$, and compare the sets of morphisms $\mathrm{hom}_{\mathbf{C}}(C, A)$ and $\mathrm{hom}_{\mathbf{C}}(C, A')$. If they are always the same, then g is an isomorphism. In other words, if every object C "sees" A and A' in exactly the same way via morphisms, then A and A' are the same up to isomorphism.

Remark. Conversely, for any $g \in \mathrm{hom}_{\mathbf{C}}(A, A')$, there is a unique natural transformation in $\mathrm{Nat}(h_A, h_{A'})$, given by $g \circ -$. Hence, we obtain a bijection between $\mathrm{Nat}(h_A, h_{A'})$ and $\mathrm{hom}_{\mathbf{C}}(A, A')$. Indeed, the so-called Yoneda embedding is just a functor:

$$\mathbf{y} : \mathbf{C} \longrightarrow \mathbf{Set}^{\mathbf{C}^{op}}$$

where $\mathbf{y}$ sends each object A to the functor h_A (how every object maps to A). It sends each morphism, say $g \in \mathrm{hom}_{\mathbf{C}}(A, A')$ to the natural transformation $i \in \mathrm{Nat}(h_A, h_{A'})$. Since we just showed that $\mathrm{hom}_{\mathbf{C}}(A, A') \cong \mathrm{Nat}(h_A, h_{A'})$, $\mathbf{y}$ is fully faithful.

2.2 Representable Functors

As we just saw, for a fixed $A \in \mathrm{Obj}(\mathbf{C})$, the two functors from $\mathbf{C}$ to $\mathbf{Set}$ that capture the "arrows", namely $h_A := \mathrm{hom}_{\mathbf{C}}(-, A)$ and $h^A := \mathrm{hom}_{\mathbf{C}}(A, -)$, are special and important. We call the functors that are naturally isomorphic to them representable functors.

Definition 2.2.1. Let $F : \mathbf{C} \longrightarrow \mathbf{Set}$ be a contravariant functor. If $\exists \Omega \in \mathrm{Obj}(\mathbf{C})$ and a natural isomorphism:

$$\eta : h_{\Omega} \xrightarrow{\sim} F$$

then we call F **representable**. The covariant case is analogous.

Example 2.2.2. To see the first example, let's recall the universal property of free objects. Given a set X , [the] free object in $\mathbf{C}$ generated by X is an object $V \in \mathrm{Obj}(\mathbf{C})$ together with an inclusion map $i : X \longrightarrow U(V)$ satisfying the following: for any $A \in \mathrm{Obj}(\mathbf{C})$ and any $f \in \mathrm{hom}_{\mathbf{Set}}(X, U(A))$,

there exists a unique $\bar{f} \in \text{hom}_{\mathbf{C}}(V, A)$ such that $U(\bar{f}) \circ i = f$. That is, the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{i} & U(V) \\ f \downarrow & \swarrow U(\bar{f}) & \\ U(A) & & \end{array}$$

Now, given a fixed set X , we define a functor $F : \mathbf{C} \rightarrow \mathbf{Set}$ by $F(A) = \text{hom}_{\mathbf{Set}}(X, U(A))$. (Convince yourself that this is indeed a functor, and find $F(u)$ where u is a morphism.) Now define $\alpha : h^V(-) \Rightarrow F(-)$ where

$$\text{hom}_{\mathbf{C}}(V, A) \xrightarrow{\alpha_A} \text{hom}_{\mathbf{Set}}(X, U(A))$$

$$\bar{f} \longmapsto U(\bar{f}) \circ i$$

I claim that α is a natural isomorphism, and hence F is representable by h^V . We first show that α is indeed a natural transformation. Suppose we have a morphism $g : A \rightarrow B$; then the following diagram commutes:

$$\begin{array}{ccc} \text{hom}_{\mathbf{C}}(V, A) & \xrightarrow{\alpha_A} & \text{hom}_{\mathbf{Set}}(X, U(A)) \\ \downarrow & & \downarrow \\ \text{hom}_{\mathbf{C}}(V, B) & \xrightarrow{\alpha_B} & \text{hom}_{\mathbf{Set}}(X, U(B)) \end{array}$$

$$\begin{array}{ccc} \bar{f} & \longmapsto & U(\bar{f}) \circ i \\ \downarrow & & \downarrow \\ g \circ \bar{f} & \longmapsto & U(g \circ \bar{f}) \circ i = U(g) \circ U(\bar{f}) \circ i \end{array}$$

Then we show that for any $A \in \text{Obj}(\mathbf{C})$, the map α_A is an isomorphism, i.e., a bijection. Indeed, surjectivity follows from the existence of $\bar{f}$, and injectivity follows from the uniqueness of $\bar{f}$.

Example 2.2.3. To see the second example, let's first recall the universal property of tensor products. Fix $A, B \in \text{Obj}(\mathbf{Mod}_R)$, [the] tensor product $A \otimes_R B$, is an object in $\mathbf{Mod}_R$, together with a canonical bilinear map $i : A \times B \rightarrow A \otimes_R B$ satisfying the following: for any $C \in \text{Obj}(\mathbf{Mod}_R)$ and any bilinear map $h : A \times B \rightarrow C$, $\exists! \bar{h} \in \text{Hom}_R(A \otimes_R B, C)$ s.t. $h = \bar{h} \circ i$. That is, the following diagram commutes:

$$\begin{array}{ccc} A \times B & \xrightarrow{h} & C \\ i \downarrow & \searrow \bar{h} & \\ A \otimes_R B & & \end{array}$$

Now, given $A, B \in \text{Obj}(\mathbf{Mod}_R)$, we define a functor $F : \mathbf{Mod}_R \rightarrow \mathbf{Set}$ by $F(C) = \text{Bil}_R(A \times B, C)$. (Again, convince yourself that this is indeed a functor, and find $F(u)$ where u is a morphism.)

Now define $\alpha : h^{A \otimes_R B}(-) \Rightarrow F(-)$ where

$$\mathrm{Hom}_R(A \otimes_R B, C) \xrightarrow{\alpha_C} \mathrm{Bil}_R(A \times B, C)$$

$$\bar{h} \longmapsto \bar{h} \circ i = h$$

Again, I claim that α is a natural isomorphism, and hence F is representable by $h^{A \otimes_R B}$. We first show that α is indeed a natural transformation. Suppose we have a morphism $g : C \rightarrow D$; then the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Hom}_R(A \otimes_R B, C) & \xrightarrow{\alpha_C} & \mathrm{Bil}_R(A \times B, C) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_R(A \otimes_R B, D) & \xrightarrow{\alpha_D} & \mathrm{Bil}_R(A \times B, D) \end{array}$$

$$\begin{array}{ccc} \bar{h} & \longmapsto & \bar{h} \circ i \\ \downarrow & & \downarrow \\ g \circ \bar{h} & \longmapsto & (g \circ \bar{h}) \circ i = g \circ (\bar{h} \circ i) \end{array}$$

Now, again, it remains to show that each α_C is an isomorphism, which is again, immediate from our construction of $A \otimes_R B$ using the universal property.

2.3 Product Categories

Definition 2.3.1. Given categories $\mathbf{C}$ and $\mathbf{D}$, we can define a new category called the **product category** on $\mathbf{C}$ and $\mathbf{D}$, denoted $\mathbf{C} \times \mathbf{D}$. The objects are all pairs (C, D) where $C \in \mathbf{C}$ and $D \in \mathbf{D}$. The morphisms from (C, D) to (C', D') are all pairs (f, g) where $f \in \mathrm{hom}_{\mathbf{C}}(C, C')$ and $g \in \mathrm{hom}_{\mathbf{D}}(D, D')$. The composition rule is $(f_1, g_1) \circ (f_2, g_2) := (f_1 \circ f_2, g_1 \circ g_2)$.

The multi-variable functors can be viewed as the functors that start from product categories. Here are some examples:

Example 2.3.2. $\mathrm{hom}_{\mathbf{C}}(-, -)$ can be viewed as a functor $\mathbf{C}^{op} \times \mathbf{C} \rightarrow \mathbf{Set}$. It sends $(C_1, C_2) \in \mathrm{Obj}(\mathbf{C}^{op} \times \mathbf{C})$ to $\mathrm{hom}_{\mathbf{C}}(C_1, C_2) \in \mathrm{Obj}(\mathbf{Set})$, and sends $(f, g) \in \mathrm{hom}_{\mathbf{C}^{op} \times \mathbf{C}}((C_1, C_2), (C'_1, C'_2))$ to $g \circ - \circ f$.

$$\begin{array}{ccc} (C_1, C_2) & \longrightarrow & \mathrm{hom}_{\mathbf{C}}(C_1, C_2) \\ (f, g) \downarrow & & \downarrow g \circ - \circ f \\ (C'_1, C'_2) & \longrightarrow & \mathrm{hom}_{\mathbf{C}}(C'_1, C'_2) \end{array} \quad \begin{array}{ccc} C_1 & \xrightarrow{u} & C_2 \\ f \uparrow & & \downarrow g \\ C'_1 & \longrightarrow & C'_2 \end{array}$$

Example 2.3.3. $\mathcal{F}(\Omega)$ can be viewed as a functor $\mathbf{C} \times \mathbf{Set}^{\mathbf{C}} \rightarrow \mathbf{Set}$, where $\Omega \in \mathrm{Obj}(\mathbf{C})$ and $\mathcal{F}$ is a covariant functor from $\mathbf{C}$ to $\mathbf{Set}$. The morphism sending $x \in F(A)$ to $G(B)$ is given by the

following:

$$\begin{array}{ccc}
(A, F) & \longrightarrow & F(A) \\
(u, \eta) \downarrow & & \downarrow (G(u) \circ \eta_A)(x) = (\eta_B \circ F(u))(x) \\
(B, G) & \longrightarrow & G(B)
\end{array}
\qquad
\begin{array}{ccc}
F(A) & \xrightarrow{\eta_A} & G(A) \\
F(u) \downarrow & & \downarrow G(u) \\
F(B) & \xrightarrow{\eta_B} & G(B)
\end{array}$$

Example 2.3.4. $\text{Nat}(h^-, -)$ can be viewed as a functor $\mathbf{C} \times \mathbf{Set}^{\mathbf{C}} \longrightarrow \mathbf{Set}$. It sends $(A, F) \in \text{Obj}(\mathbf{C} \times \mathbf{Set}^{\mathbf{C}})$ to $\text{Nat}(h^A, F) \in \text{Obj}(\mathbf{Set})$, and send (u, η) to $\eta \circ - \circ \xi$, where ξ is the unique natural transformation $\in \text{Nat}(h^B, h^A)$ induced by $u \in \text{hom}_{\mathbf{C}}(A, B)$, due to the Yoneda embedding.

$$\begin{array}{ccccc}
(A, F) & \longrightarrow & \text{Nat}(h^A, F) & h^A \xRightarrow{\theta} F & \text{hom}_{\mathbf{C}}(B, \Omega) \xrightarrow{\xi_{\Omega} = - \circ u} \text{hom}_{\mathbf{C}}(A, \Omega) \\
(u, \eta) \downarrow & & \downarrow \eta \circ - \circ \xi & \xi \uparrow \Downarrow \eta & \downarrow \downarrow \\
(B, G) & \longrightarrow & \text{Nat}(h^B, G) & h^B \xRightarrow{\quad} G & \text{hom}_{\mathbf{C}}(B, \Omega') \xrightarrow{\xi_{\Omega'} = - \circ u} \text{hom}_{\mathbf{C}}(A, \Omega')
\end{array}$$

2.4 Yoneda's Lemma

Here comes to the ultra version of the Yoneda's lemma. As we just saw in Example 2.3.3 and Example 2.3.4, we have two functors from $\mathbf{C} \times \mathbf{Set}^{\mathbf{C}}$ to $\mathbf{Set}$. In fact, they are naturally isomorphic. That is:

Theorem 2.4.1 (Yoneda's Lemma). *Given $F \in \text{Obj}(\mathbf{Set}^{\mathbf{C}})$, i.e. a covariant functor $\mathbf{C} \longrightarrow \mathbf{Set}$; and $A \in \text{Obj}(\mathbf{C})$, there is natural isomorphism*

$$\Phi_{A, F} : \text{Nat}(h^A, F) \xrightarrow{\sim} F(A)$$

Proof. First, we fix F , and let

$$\Phi_A : \text{Nat}(h^A, F) \longrightarrow F(A)$$

$$\eta \longmapsto \eta_A(\text{id}_A)$$

Convince yourself that Φ_A is a bijection by properly defining the inverse Ψ_A . Define

$$\Psi_A : F(A) \longrightarrow \text{Nat}(h^A, F)$$

$$x \longmapsto \Psi_A(x)$$

where $\Psi_A(x) : h^A \Rightarrow F$ is the natural transformation whose component at $C \in \text{Obj}(\mathbf{C})$ is the map

$$\Psi_A(x)_C : h^A(C) \longrightarrow F(C) \quad f \mapsto F(f)(x)$$

Then, we verify that Φ is a natural transformation in A . Let $u : A \longrightarrow B$ be a morphism in $\mathbf{C}$. By the Yoneda embedding, u induces a natural transformation $\xi \in \text{Nat}(h^B, h^A)$ whose component at

C is $\xi_C = - \circ u$. We need to check that the following diagram commutes:

$$\begin{array}{ccc}
\text{Nat}(h^A, F) & \xrightarrow{\Phi_A} & F(A) \\
\downarrow - \circ \xi & & \downarrow F(u) \\
& \begin{array}{ccc} \eta & \xrightarrow{\quad} & \eta_A(\text{id}_A) \\ \downarrow & & \downarrow \\ \eta \circ \xi & \xrightarrow{\quad} & \dots \end{array} & \\
\text{Nat}(h^B, F) & \xrightarrow{\Phi_B} & F(B)
\end{array}$$

Take $\eta \in \text{Nat}(h^A, F)$. The bottom-left route gives

$$\Phi_B(\eta \circ \xi) = (\eta \circ \xi)_B(\text{id}_B) = \eta_B(\xi_B(\text{id}_B)) = \eta_B(\text{id}_B \circ u) = \eta_B(u)$$

The top-right route gives

$$F(u)(\eta_A(\text{id}_A))$$

But note that we have the naturality of η :

$$\begin{array}{ccc}
\text{hom}_{\mathbf{C}}(A, A) & \xrightarrow{\eta_A} & F(A) \\
\downarrow u \circ - & & \downarrow F(u) \\
& \begin{array}{ccc} \text{id}_A & \xrightarrow{\quad} & \eta_A(\text{id}_A) \\ \downarrow & & \downarrow \\ u & \xrightarrow{\quad} & \eta_B(u) = F(u)(\eta_A(\text{id}_A)) \end{array} & \\
\text{hom}_{\mathbf{C}}(A, B) & \xrightarrow{\eta_B} & F(B)
\end{array}$$

So, we are done.

Second, we fix A , and let

$$\Phi_F : \text{Nat}(h^A, F) \longrightarrow F(A)$$

$$\eta \longmapsto \eta_A(\text{id}_A)$$

We have shown that this is a bijection, so it remains to show the naturality of Φ . Indeed, let $\theta \in \text{Nat}(F, G)$ be any natural transformation where $F, G \in \text{Obj}(\mathbf{Set}^{\mathbf{C}})$. The following diagram

commutes:

$$\begin{array}{ccc}
\text{Nat}(h^A, F) & \xrightarrow{\Phi_F} & F(A) \\
\downarrow \theta \circ - & & \downarrow \theta_A \\
\text{Nat}(h^A, G) & \xrightarrow{\Phi_G} & G(A)
\end{array}$$

$$\begin{array}{ccc}
\eta & \xrightarrow{\quad} & \eta_A(\text{id}_A) \\
\downarrow & & \downarrow \\
\theta \circ \eta & \xrightarrow{\quad} & \dots
\end{array}$$

Take $\eta \in \text{Nat}(h^A, F)$. The bottom-left route gives

$$\Phi_G(\theta \circ \eta) = (\theta \circ \eta)_A(\text{id}_A) = \theta_A(\eta_A(\text{id}_A))$$

which is exactly the result of the top-right route, as desired. $\square$

Remark. It is worth mentioning that if we take $F = h^{A'}$ (covariant case) or $F = h_{A'}$ (contravariant case), then Yoneda's lemma reduces to the Yoneda embedding.

Here we give two applications (examples) of the Yoneda embedding. I follow the statements in [3].

Corollary 2.4.2. *Every row operation on matrices with n rows is defined by left multiplication by some $n \times n$ matrix, namely the matrix obtained by performing the row operation on the identity matrix.*

Proof. We define a category $\mathbf{Mat}_R$ where the objects are positive integers and the morphisms $\text{hom}(m, n)$ are all the $n \times m$ matrices whose entries are elements in R . As for composition of morphisms, let $\phi \in \text{hom}(m, n)$ and $\psi \in \text{hom}(n, p)$; we define $\psi \circ \phi := \psi \cdot \phi$, the product of the matrices ψ and ϕ .

Now fix a positive integer n as the number of rows. We may ask: what is a row operation on matrices with n rows? Note that the functor $h_n = \text{hom}(-, n)$, viewed as a set, is the set of "all the matrices with n rows over R ". Hence, given a row operation η , we may think of it as a natural transformation $h_n \Rightarrow h_n$. Indeed, fix arbitrary positive integers m_1, m_2 and an arbitrary $m_1 \times m_2$ matrix over R , say $U \in \text{hom}(m_2, m_1)$; we have:

$$\begin{array}{ccc}
\text{hom}(m_1, n) & \xrightarrow{\eta_{m_1}} & \text{hom}(m_1, n) \\
\downarrow - \cdot U & & \downarrow - \cdot U \\
\text{hom}(m_2, n) & \xrightarrow{\eta_{m_2}} & \text{hom}(m_2, n)
\end{array}$$

This diagram commutes because we can switch the order of right multiplication and row operation

(why? convince yourself). Therefore, by the Yoneda embedding, we have the following bijection:

$$\Phi : \text{Nat}(h_n, h_n) \longrightarrow \text{hom}(n, n)$$

$$\eta \longmapsto \eta_n(\text{id}_n)$$

$$V \circ - \longleftarrow \longrightarrow V$$

$V \circ -$ is just left multiplication by the matrix V , hence every row operation must be defined by left multiplication by some $n \times n$ matrix. What is this matrix? Note that id_n is the $n \times n$ identity matrix $\mathbf{I}_{n \times n}$. Hence, it is obtained by applying the corresponding row operation to $\mathbf{I}_{n \times n}$. $\square$

Corollary 2.4.3 (Cayley). *Any group G is isomorphic to a subgroup of $\text{Sym}(G)$, the group of all bijections from G to G , i.e. $\text{Aut}_{\text{Set}}(G)$.*

We postpone the proof to later and do some preparation first. Given a group G , let's make it into a category $\mathbf{BG}$ where the only object is an abstract element $*$ and the morphisms are the elements of G . Consider a covariant functor:

$$F : \mathbf{BG} \longrightarrow \mathbf{C}$$

where $F(*) =: X \in \text{Obj}(\mathbf{C})$, which can be interpreted as different things depending on the category $\mathbf{C}$: X is called a **left G -set** / **G -representation** / **G -space** if $\mathbf{C} = \mathbf{Set}$ / $\mathbf{Vect}_{\mathbb{K}}$ / $\mathbf{Top}$. If F is contravariant and $\mathbf{C} = \mathbf{Set}$, then X is called a **right G -set**.

Example 2.4.4. Let $\mathbf{C} = \mathbf{Set}$ and fix a covariant functor $F : \mathbf{BG} \longrightarrow \mathbf{Set}$, then F determines a left group action by the following: take $g \in \text{hom}_{\mathbf{BG}}(*, *)$, $F(g)$ is a map $X \longrightarrow X$. The action is given by:

$$\phi : G \times X \longrightarrow X$$

$$(g, x) \longmapsto F(g)(x)$$

Example 2.4.5. Let $\mathbf{C} = \mathbf{Vect}_{\mathbb{K}}$ and fix a covariant functor $F : \mathbf{BG} \longrightarrow \mathbf{Vect}_{\mathbb{K}}$. Let $V := F(*) \in \text{Obj}(\mathbf{Vect}_{\mathbb{K}})$, then F determines a G -representation by the following:

$$\rho : G \longrightarrow \text{GL}(V)$$

$$g \longmapsto F(g)$$

Note that since every morphism in $\mathbf{BG}$ is invertible, $F(g)$ is an isomorphism in $\mathbf{Vect}_{\mathbb{K}}$, hence an element of $\text{GL}(V)$.

Definition 2.4.6. Let A and B be left G -sets (with corresponding G -actions). A morphism $f : A \longrightarrow B$ is said to be **left G -equivariant** if for any $g \in G$ and any $a \in A$, $f(g \bullet a) = g \bullet f(a)$. That is, the following diagram commutes:

$$\begin{array}{ccc} A & \xrightarrow{f(-)} & B \\ g \bullet - \downarrow & & \downarrow g \bullet - \\ A & \xrightarrow{f(-)} & B \end{array}$$

In other words, f does not break the structure of the G -actions. The definition of right G -equivariant follows analogously.

Lemma 2.4.7. *Let G be regarded as a right G -set via right multiplication. Then for any right G -set X ,*

$$\mathrm{hom}_{G\text{-}\mathbf{Set}}(G, X) \cong X$$

via $\phi \mapsto \phi(e)$.

Proof. A right G -set X corresponds to a functor $F : \mathbf{BG}^{op} \rightarrow \mathbf{Set}$ with $F(*) = X$. The representable functor $h_* = \mathrm{hom}_{\mathbf{BG}}(-, *)$ corresponds to the right G -set G . Note that in this setting, we have a one-to-one correspondence:

$$\{\text{right } G\text{-equivariant maps} \in \mathrm{hom}_{G\text{-}\mathbf{Set}}\} \longleftrightarrow \{\text{natural transformations} \in \mathrm{Nat}(h_*, F)\}$$

That is

$$\mathrm{hom}_{G\text{-}\mathbf{Set}}(G, X) \cong \mathrm{Nat}(h_*, F)$$

Then, by Yoneda's lemma, we have the following natural isomorphism:

$$\Phi : \mathrm{Nat}(h_*, F) \xrightarrow{\sim} F(*)$$

$$\phi \longmapsto \phi(e)$$

But since $F(*) = X$, we obtain $\mathrm{hom}_{G\text{-}\mathbf{Set}}(G, X) \cong X$, as desired. $\square$

Proof of the Cayley's theorem. Let X be G itself, then the above natural isomorphism reduces to the Yoneda embedding:

$$\Phi : \mathrm{Nat}(h_*, h_*) \xrightarrow{\sim} \mathrm{hom}_{\mathbf{BG}}(*, *) = G$$

$$g \circ - \longleftarrow \longmapsto g$$

where $g \circ -$ is just the left multiplication. This means every right G -equivariant map in $\mathrm{hom}_{G\text{-}\mathbf{Set}}(G, G)$ is defined by a left multiplication by some $g \in G$, hence invertible. As a consequence, every G -equivariant endomorphism of G must be an automorphism! And:

$$\mathrm{Aut}_{G\text{-}\mathbf{Set}}(G) \cong G$$

Now, we composite the Yoneda embedding and the forgetful functor, which are both faithful:

$$\mathbf{BG} \xhookrightarrow{\quad} \mathbf{Set}^{\mathbf{BG}^{op}} \xhookrightarrow{U} \mathbf{Set}$$

$$\text{morphsim } G \xhookrightarrow{\cong} \mathrm{Aut}_{G\text{-}\mathbf{Set}}(G) \hookrightarrow \mathrm{Sym}(G)$$

and obtain:

$$G \hookrightarrow \mathrm{Sym}(G)$$

as desired. $\square$

2.5 Adjoint Functors

Starting from this section, [4] will be the main reference. When the context is clear, I will omit the notation $\text{Obj}(-)$. If not otherwise specified, A and B will denote commutative rings with unity.

Just as universal property "essentially" determines an object in a category (assuming existence), adjoints "essentially" determine a functor (assuming existence). Let $F : \mathbf{C} \longrightarrow \mathbf{D}$ and $G : \mathbf{D} \longrightarrow \mathbf{C}$ be covariant functors, then $\text{hom}_{\mathbf{C}}(-, G(-))$ and $\text{hom}_{\mathbf{D}}(F(-), -)$ can be viewed as functors

$$\mathbf{C}^{op} \times \mathbf{D} \longrightarrow \mathbf{Set}$$

A natural transformation τ between these two functors is just: $\forall C \in \mathbf{C}, D \in \mathbf{D}, \exists$ bijection:

$$\tau_{C,D} : \text{hom}_{\mathbf{C}}(C, G(D)) \xrightarrow{\cong} \text{hom}_{\mathbf{D}}(F(C), D)$$

s.t. for every pair of morphisms $u \in \text{hom}_{\mathbf{C}}(C', C), v \in \text{hom}_{\mathbf{D}}(D, D')$, the following diagram commutes:

$$\begin{array}{ccc} \text{hom}_{\mathbf{C}}(C, G(D)) & \xrightarrow{\tau_{C,D}} & \text{hom}_{\mathbf{D}}(F(C), D) \\ \downarrow G(v) \circ - \circ u & & \downarrow v \circ - \circ F(u) \\ \text{hom}_{\mathbf{C}}(C', G(D')) & \xrightarrow{\tau_{C',D'}} & \text{hom}_{\mathbf{D}}(F(C'), D') \end{array}$$

Definition 2.5.1. We say that (F, G) forms an **adjoint pair**, and that F is **left-adjoint** to G and G is **right-adjoint** to F . We also say that F is a **left adjoint** and G is a **right adjoint**.

Example 2.5.2. Let $M, N, P \in \mathbf{Mod}_A$. We have

$$\text{Hom}_A(M \otimes_A N, P) \xrightarrow{\cong} \text{Bil}_A(M \times N, P) \xrightarrow{\phi} \text{Hom}_A(M, \text{Hom}_A(N, P))$$

$$\bar{h} \longmapsto h \longmapsto \tilde{h} : [m \mapsto h(m, -)]$$

It easy to check that ϕ is an isomorphism by defining $\psi :$

$$\psi : \text{Hom}_A(M, \text{Hom}_A(N, P)) \longrightarrow \text{Bil}_A(M \times N, P)$$

$$\tilde{g} \longmapsto g : [(m, n) \mapsto \tilde{g}(m)(n)]$$

Now fix N . I claim that $F(-) := - \otimes_A N$ and $G(-) := \text{Hom}_A(N, -)$ are adjoint functors. Let $u \in \text{Hom}_A(M_2, M_1), v \in \text{Hom}_A(P_1, P_2)$. So $F(u)(-) = (u \otimes_A \text{id}_N)(-)$ and $G(v)(-) = v \circ -$.

$$\begin{array}{ccc} \text{Hom}_A(M_1, \text{Hom}_A(N, P_1)) & \xrightarrow{\eta_{M_1, P_1}} & \text{Hom}_A(M_1 \otimes_A N, P_1) \\ \downarrow & & \downarrow \\ \tilde{g} \longmapsto \bar{g} & & \\ \downarrow & & \downarrow \\ v \circ \tilde{g} \circ u \xrightarrow{?} v \circ \bar{g} \circ (u \otimes_A \text{id}_N) & & \\ \downarrow & & \downarrow \\ \text{Hom}_A(M_2, \text{Hom}_A(N, P_2)) & \xrightarrow{\eta_{M_2, P_2}} & \text{Hom}_A(M_2 \otimes_A N, P_2) \end{array}$$

where

$$\eta_{M,P} : \text{Hom}_A(M, \text{Hom}_A(N, P)) \longrightarrow \text{Hom}_A(M \otimes_A N, P)$$

$$\tilde{g} \longmapsto \bar{g} : [m \otimes_A n \mapsto \tilde{g}(m)(n)]$$

Now, let's check:

$$\text{Bottom-Left: } \eta_{M_2, P_2}(v \circ \tilde{g} \circ u) : m \otimes_A n \mapsto (v \circ \tilde{g} \circ u)(m)(n) = v(\tilde{g}(u(m)))(n) = v(f(n))$$

$$\text{Upper-Right: } v \circ \bar{g} \circ (u \otimes_A \text{id}_N) : m \otimes_A n \mapsto v(\bar{g}(u(m) \otimes_A n)) = v(\tilde{g}(u(m))(n)) = v(f(n))$$

where $f := \tilde{g}(u(m)) \in \text{Hom}_A(N, P_1)$. Done.

Example 2.5.3. Let $M \in \mathbf{Mod}_A$; $f : B \longrightarrow A$ a ring homomorphism. Recall that we can regard M as a B -module M_B by

$$b \bullet m := f(b) \bullet m$$

This gives a functor $G : \mathbf{Mod}_A \longrightarrow \mathbf{Mod}_B$:

$$\begin{array}{ccc} M_1 & \xrightarrow{G(-)} & M_{1B} \\ v \downarrow & & \downarrow v \\ M_2 & \xrightarrow{G(-)} & M_{2B} \end{array}$$

Now I claim that $G(-)$ is right adjoint to $F(-) : - \otimes_B A : \mathbf{Mod}_B \longrightarrow \mathbf{Mod}_A$:

$$\begin{array}{ccc} N_2 & \xrightarrow{F(-)} & N_2 \otimes_B A \\ u \downarrow & & \downarrow u \otimes_B \text{id}_A \\ N_1 & \xrightarrow{F(-)} & N_1 \otimes_B A \end{array}$$

First, we construct the bijection:

$$\Phi : \text{Hom}_B(N, M_B) \longrightarrow \text{Hom}_A(N \otimes_B A, M)$$

$$g \longmapsto [n \otimes_B a \mapsto a \bullet g(n)]$$

$$\Psi : \text{Hom}_A(N \otimes_B A, M) \longrightarrow \text{Hom}_B(N, M_B)$$

$$h \longmapsto [n \mapsto h(n \otimes_B 1_A)]$$

We need to check that for each $g \in \text{Hom}_B(N, M_B)$, the assignment $\Phi(g)$ is well-defined and A -linear. Indeed, for $b \in B$ we have

$$\begin{aligned} \Phi(g)(b \bullet n \otimes_B a) &= a \bullet g(b \bullet n) = a \bullet (b \bullet g(n)) = a \bullet (f(b) \bullet g(n)) = (f(b)a) \bullet g(n) \\ &= \Phi(g)(n \otimes_B f(b)a) \end{aligned}$$

so $\Phi(g)$ respects the B -balanced relation in $N \otimes_B A$, hence is well defined. Moreover, for $a' \in A$,

$$\Phi(g)((n \otimes_B a)a') = \Phi(g)(n \otimes_B aa') = (aa') \bullet g(n) = a' \bullet (a \bullet g(n)),$$

hence $\Phi(g)$ is A -linear. Therefore $\Phi(g) \in \text{Hom}_A(N \otimes_B A, M)$.

Now, since h is A -linear (for the sake of the second line), we have:

$$\begin{aligned} (\Psi \circ \Phi)(g) : n &\mapsto \Phi(g)(n \otimes_B 1_A) = 1_A \bullet g(n) = g(n) \\ (\Phi \circ \Psi)(h) : n \otimes_B a &\mapsto a \bullet (\Psi(h)(n)) = a \bullet h(n \otimes_B 1_A) = h(n \otimes_B a) \end{aligned}$$

Therefore, both $\Psi \circ \Phi$ and $\Phi \circ \Psi$ are identical maps. Finally, we verify the naturality:

$$\begin{array}{ccc} \text{Hom}_B(N_1, M_{1B}) & \xrightarrow{\Phi_{N_1, M_1}} & \text{Hom}_A(N_1 \otimes_B A, M_1) \\ \downarrow & & \downarrow \\ & \begin{array}{ccc} g & \xrightarrow{\quad} & \Phi(g) \\ \downarrow & & \downarrow \\ v \circ g \circ u & \xrightarrow{?} & v \circ \Phi(g) \circ (u \otimes_B \text{id}_A) \end{array} & \\ \downarrow & & \downarrow \\ \text{Hom}_B(N_2, M_{2B}) & \xrightarrow{\Phi_{N_2, M_2}} & \text{Hom}_A(N_2 \otimes_B A, M_2) \end{array}$$

So it suffices to show: $\Phi(v \circ g \circ u) = v \circ \Phi(g) \circ (u \otimes_B \text{id}_A)$. Indeed, since v is A -linear, we have:

$$\begin{aligned} \Phi(v \circ g \circ u) : n \otimes_B a &\mapsto a \bullet (v \circ g \circ u)(n) = v(a \bullet g(u(n))) \\ v \circ \Phi(g) \circ (u \otimes_B \text{id}_A) : n \otimes_B a &\mapsto v(\Phi(g)(u(n) \otimes_B a)) = v(a \bullet g(u(n))) \end{aligned}$$

In fact, the following theorem tells us that the functor $G(-)$ in the example above has to have a left adjoint.

Theorem 2.5.4. *A covariant functor $G : \mathbf{D} \longrightarrow \mathbf{C}$ has a left adjoint if and only if for any $C \in \mathbf{C}$, the functor $\text{hom}_{\mathbf{C}}(C, G(-))$ is representable.*

In the context of the above example, we have $\mathbf{C} = \mathbf{Mod}_B$, $\mathbf{D} = \mathbf{Mod}_A$, and $G(M) = M_B$. Hence for any fixed $N \in \mathbf{Mod}_B$, the functor

$$\text{hom}_{\mathbf{C}}(N, G(-)) = \text{Hom}_B(N, (-)_B)$$

is represented by $N \otimes_B A$, since for any $M \in \mathbf{Mod}_A$ we have a natural bijection

$$\text{Hom}_B(N, M_B) \xrightarrow{\cong} \text{Hom}_A(N \otimes_B A, M) = h^{N \otimes_B A}(M)$$

Proof of Theorem 2.5.4. $\implies$: Let $F : \mathbf{C} \longrightarrow \mathbf{D}$ be a left adjoint of G . Then for any pair $(C, D) \in \mathbf{C} \times \mathbf{D}$, we have a bijection

$$\tau_{C,D} : \text{hom}_{\mathbf{C}}(C, G(D)) \xrightarrow{\cong} \text{hom}_{\mathbf{D}}(F(C), D)$$

This map is natural with respect to both C and D . Hence if we fix C , we obtain a natural isomorphism of functors $\mathbf{D} \rightarrow \mathbf{Set}$:

$$\tau_{C,-} : \text{hom}_{\mathbf{C}}(C, G(-)) \xrightarrow{\sim} h^{F(C)}(-)$$

Therefore $\text{hom}_{\mathbf{C}}(C, G(-))$ is represented by $F(C)$.

$\Leftarrow$: Suppose that for each $C \in \mathbf{C}$, the functor $\text{hom}_{\mathbf{C}}(C, G(-))$ is representable. Let $A_C \in \mathbf{D}$ be a representing object. Define a functor $F : \mathbf{C} \rightarrow \mathbf{D}$ by

$$F(C) = A_C.$$

Then by the definition of representability, we have a natural isomorphism

$$\text{hom}_{\mathbf{D}}(F(C), -) \xrightarrow{\sim} \text{hom}_{\mathbf{C}}(C, G(-)).$$

Using the universal property of representing objects, one verifies that this assignment extends to a functor F and that the above bijections are natural in both variables. We omit the detailed verification. $\square$

Here is another motivating example:

Example 2.5.5 ((Abelian) Groupification of abelian semigroups). An **abelian semigroup** is just like an abelian group, except we do not require an identity or an inverse. Morphisms are maps of sets preserving the binary operation. One example is the non-negative integers under addition. Another example is the positive integers under multiplication. A **abelian groupification** of an abelian semigroup $S \in \mathbf{SAb}$ is a map $\pi : S \rightarrow G$ s.t. $G \in \mathbf{Ab}$, and any map of abelian semigroups from S to an abelian group $G' \in \mathbf{Ab}$ factors uniquely through G :

$$\begin{array}{ccc} S & \xrightarrow{\pi} & G \\ \downarrow & \swarrow \exists! & \\ U(G') & & \end{array}$$

Now, we may construct a functor $H : \mathbf{SAb} \rightarrow \mathbf{Ab}$ as follows: for an abelian semigroup $S \in \mathbf{SAb}$, $H(S) \in \mathbf{Ab}$ gives an abelian group whose underlying set is $S \times S$, where the elements are ordered pairs $(a, b) \in S \times S$ (mod out some relation), which you may think of as $a - b$ (We need to introduce subtraction!), with the equivalence that

$$(a, b) \sim (c, d) \iff a + d + e = b + c + e \text{ for some } e \in S$$

Convince yourself that $\sim$ is indeed an equivalence relation. Now define addition on $S \times S / \sim$ as follows:

$$[(a, b)] + [(c, d)] := [(a + c, b + d)]$$

Check the well-defineness: Suppose $(a', b') \sim (a, b)$ and $(c', d') \sim (c, d)$. i.e. $a' + b + e = a + b' + e$ and $c' + d + f = c + d' + f$ for some $e, f \in S$. Then

$$[(a', b')] + [(c', d')] = [(a' + c', b' + d')]$$

Note that

$$a' + c' + b + d + e + f = a + c + b' + d' + e + f$$

Hence $(a' + c', b' + d') \sim (a + c, b + d)$, as we expected.

Then the identity of $H(S) = S \times S / \sim$, as we may guess, is:

$$0_{H(S)} = [(a, a)], \quad \forall a \in S$$

The inverse, as we may guess, is: $-[(a, b)] := [(b, a)]$. Therefore, $H(S)$ with the rules defined above, forms an abelian group. Now, in order to make H a functor, we need to specify $H(u)$ where $u \in \text{hom}_{\mathbf{SAb}}(S, T)$. Define

$$H(u) : S \times S / \sim \longrightarrow T \times T / \sim$$

$$[(a, b)] \longmapsto [(u(a), u(b))]$$

It's straightforward to verify that H is indeed a functor. Now, it remains to show that H is indeed an abelian groupification, i.e. $H(S)$, together with π , satisfy the universal property.....

$$\begin{array}{ccc} S & \xrightarrow{\pi} & H(S) \\ \downarrow f \in \text{hom}_{\mathbf{SAb}}(S, U(G')) & \swarrow \exists! \bar{f} \in \text{hom}_{\mathbf{Ab}}(H(S), G') & \\ U(G') & & \end{array}$$

You may wish to define π as $s \mapsto [(s, 0_S)]$. However, S does not necessarily have an identity! So instead, we define $\pi : s \mapsto [(s + s, s)]$. Convince yourself that π is indeed a morphism in $\mathbf{SAb}$. Now, given $f \in \text{hom}_{\mathbf{SAb}}(S, G')$, we define:

$$\bar{f} : [(a, b)] \mapsto f(a) - f(b)$$

First, check the well-defineness: Suppose $(a, b) \sim (a', b')$. Then $\exists e \in S$ s.t. $a + b' + e = a' + b + e$. Apply f on both sides:

$$f(a + b' + e) = f(a) + f(b') + f(e) = f(a') + f(b) + f(e) = f(a' + b + e)$$

Note that we can cancel $f(e)$ on both sides since we are in a group now. Hence $f(a) - f(b) = f(a') - f(b')$, as we desired.

Then, verify that $\bar{f}$ is indeed a morphism in $\mathbf{Ab}$ (omitted). Verify that $f = \bar{f} \circ \pi$ (omitted).

Finally, verify the uniqueness of $\bar{f}$: Suppose that $g \in \text{hom}_{\mathbf{Ab}}(H(S), G')$ satisfies $f = g \circ \pi$. A key observation: $[(a, b)] = \pi(a) - \pi(b)$ (Why? Convince yourself). Hence

$$g([(a, b)]) = g(\pi(a) - \pi(b)) = g(\pi(a)) - g(\pi(b)) = f(a) - f(b) = \bar{f}([(a, b)])$$

So $g = \bar{f}$, as we desired.

Now, I claim that $H(-) : \mathbf{SAb} \longrightarrow \mathbf{Ab}$ is left adjoint to the forgetful functor $U(-) : \mathbf{Ab} \longrightarrow \mathbf{SAb}$. That is, we have a natural isomorphism:

$$\mathrm{hom}_{\mathbf{SAb}}(S, U(G)) \xrightarrow{\sim} \mathrm{hom}_{\mathbf{Ab}}(H(S), G)$$

To prove this, first we construct the isomorphism

$$\tau_{S,G} : \mathrm{hom}_{\mathbf{SAb}}(S, U(G)) \longrightarrow \mathrm{hom}_{\mathbf{Ab}}(H(S), G)$$

$$f \longmapsto \bar{f}$$

$$[s \mapsto g([(s + s, s)])] \longleftarrow g$$

Left $\mapsto$ right $\mapsto$ left:

$$s \mapsto \bar{f}([(s + s, s)]) = f(s + s) - f(s) = f(s) + f(s) - f(s) = f(s)$$

Right $\mapsto$ left $\mapsto$ right:

$$\begin{aligned} [(a, b)] \mapsto f(a) - f(b) &= g([(a + a, a)]) - g([(b + b, b)]) \\ &= g([(a + a, a)] - [(b + b, b)]) \\ &= g([a, b]) \end{aligned}$$

Then, verify the naturality (omitted).

Remark. As we may notice, generally:

$$\mathrm{hom}_{\mathbf{Set}}(X, U(-)) \xrightarrow{\sim} \mathrm{hom}_{\mathbf{C}}(F(X), -)$$

where $F(X)$ is the free object generated by X . In other words:

Freedom is the left adjoint of forgetting.

The next example gives another example of "adjoints of forgetful functors", this means by which we get group from semigroups, and sheaves from presheaves. Before that, we need more definitions:

Definition 2.5.6. A **subcategory** $\mathbf{C}$ of a category $\mathbf{D}$ has as its objects some of the object of $\mathbf{D}$, and some of the morphisms, such that the morphisms of $\mathbf{C}$ include the identity morphisms of the objects of $\mathbf{C}$, and are closed under composition. Note that we have an obvious "inclusion functor" $i : \mathbf{C} \longrightarrow \mathbf{D}$. A subcategory $i : \mathbf{C} \longrightarrow \mathbf{D}$ is a **full subcategory** if i is full. That is, a subcategory $\mathbf{C}'$ of $\mathbf{C}$ is full if and only if $\forall A, B \in \mathbf{C}', \mathrm{hom}_{\mathbf{C}'}(A, B) = \mathrm{hom}_{\mathbf{C}}(A, B)$. For example, the category of f.g. A -modules is a full subcategory of $\mathbf{Mod}_A$.

Example 2.5.7. $\mathbf{Mod}_{S^{-1}A}$ is a full subcategory of $\mathbf{Mod}_A$. Then $F : \mathbf{Mod}_A \longrightarrow \mathbf{Mod}_{S^{-1}A}$ can be interpreted as adjoint to the forgetful functor $U : \mathbf{Mod}_{S^{-1}A} \longrightarrow \mathbf{Mod}_A$. The accurate constructions are as follows:

Recall: given a A -module M , we want to make it into a $S^{-1}A$ -module. Define an equivalence relation $\sim$ on $M \times S$:

$$(m, s) \sim (m', s') \iff t \bullet (s \bullet m' - s' \bullet m) = 0 \text{ for some } t \in S$$

Let $\frac{m}{s}$ denote $[(m, s)]$. Let $S^{-1}M$ (or $F(M)$) denote $M \times S / \sim$. Verify that this is indeed a $S^{-1}A$ -module with the obvious addition and ring action. Namely:

$$\frac{a}{s} \bullet \frac{m}{s'} := \frac{a \bullet m}{ss'} \quad \frac{m_1}{s_1} + \frac{m_2}{s_2} := \frac{s_2 \bullet m_1 + s_1 \bullet m_2}{s_1 s_2}$$

Now, let $u \in \text{hom}_{\mathbf{Mod}_A}(M, N)$. To make F into a functor, we need to specify $F(u)$:

$$F(u) : S^{-1}M \longrightarrow S^{-1}N$$

$$[(m, s)] = \frac{m}{s} \longmapsto [(u(m), s)] = \frac{u(m)}{s}$$

Verify that $F(u)$ is a well-defined morphism and $F(u \circ v) = F(u) \circ F(v)$. Hence $F : \mathbf{Mod}_A \longrightarrow \mathbf{Mod}_{S^{-1}A}$ is indeed a functor. Now, define $\iota : M \longrightarrow S^{-1}M$ by $\iota(m) = \frac{m}{1_A}$.

We also need to define the forgetful functor $U : \mathbf{Mod}_{S^{-1}A} \longrightarrow \mathbf{Mod}_A$ in this context. Let $Q \in \mathbf{Mod}_{S^{-1}A}$. Define $U(Q) := Q \in \mathbf{Mod}_A$ with A -action: $a \bullet q := \frac{a}{1_A} \bullet q$. Let u be a morphism in $\mathbf{Mod}_{S^{-1}A}$. Define $U(u) = u$. To show that this is well-defined, we need to verify that u is an A -linear map. Indeed:

$$a \bullet u(q) = \frac{a}{1_A} \bullet u(q) = u\left(\frac{a}{1_A} \bullet q\right) = u(a \bullet q)$$

Now, I claim that $F(M)$ satisfies the following universal property: Any A -linear map f from M to $U(Q) \in \mathbf{Mod}_A$ factors uniquely through $S^{-1}M$:

$$\begin{array}{ccc} M & \xrightarrow{\iota} & F(M) \\ \downarrow f \in \text{Hom}_A(M, U(Q)) & \swarrow \exists! \bar{f} \in \text{Hom}_{S^{-1}A}(F(M), Q) & \\ U(Q) & & \end{array}$$

To show this, first we need to define $\bar{f}$. Note that Q is an arbitrary $S^{-1}A$ -module, not necessarily in the form of $S^{-1}N$! Define:

$$\bar{f} : F(M) = S^{-1}M \longrightarrow Q$$

$$\frac{m}{s} \longmapsto s^{-1} \bullet f(m)$$

Convince yourself that $\bar{f}$ is well-defined and $S^{-1}A$ -linear. Clearly $f = \bar{f} \circ \iota$. To show uniqueness, if g also satisfies $f = g \circ \iota$, then

$$g\left(\frac{m}{s}\right) = g\left(s^{-1} \bullet \iota(m)\right) = s^{-1} \bullet g(\iota(m)) = s^{-1} \bullet f(m) = \bar{f}\left(\frac{m}{s}\right)$$

so $g = \bar{f}$. Therefore, as before, we have

$$\text{Hom}_A(M, U(Q)) \xrightarrow{\sim} \text{Hom}_{S^{-1}A}(S^{-1}M, Q)$$

2.6 Limits and Colimits as Functors

Definition 2.6.1. Given a small index category $\mathbf{J}$ and an arbitrary category $\mathbf{C}$, we may construct the **functor category** $\mathbf{C}^{\mathbf{J}}$, where the objects are the functors from $\mathbf{J}$ to $\mathbf{C}$, which are precisely the diagrams indexed by $\mathbf{J}$. For each object A in $\mathbf{C}$, we may define a **constant diagram** $\Delta_A : \mathbf{J} \rightarrow \mathbf{C}$ in $\text{Obj}(\mathbf{C}^{\mathbf{J}})$ that assigns each $j \in \text{Obj}(\mathbf{J})$ to A and every morphism in $\mathbf{J}$ to id_A . Then, we may define the so-called **diagonal functor**:

$$\begin{aligned}\Delta : \mathbf{C} &\longrightarrow \mathbf{C}^{\mathbf{J}} \\ A &\longmapsto \Delta_A\end{aligned}$$

And for each morphism $f : A \rightarrow B$, Δ sends f to the natural transformation η in $\text{Nat}(\Delta_A, \Delta_B)$, given by $\eta_j = f$ for every $j \in \text{Obj}(\mathbf{J})$. That is, the following diagram commutes:

$$\begin{array}{ccc}\Delta_A(k) = A & \xrightarrow{\eta_k = f} & \Delta_B(k) = B \\ \text{id}_A \downarrow & & \downarrow \text{id}_B \\ \Delta_A(\ell) = A & \xrightarrow{\eta_\ell = f} & \Delta_B(\ell) = B\end{array}$$

Diagonal functors provide a way to define limits and colimits of diagrams:

Recall that in the definition 1.2.1, given a diagram (i.e., a functor) $F : \mathbf{J} \rightarrow \mathbf{C}$, a limit is nothing but an object, say $\lim F$ in $\mathbf{C}$, together with a family of morphisms $\theta_i : \lim F \rightarrow F(i)$ satisfying the universal property:

$$\begin{aligned}\text{Given any object } X \text{ in } \mathbf{C}, \text{ together with a family of compatible morphisms } \xi_i : \\ X \rightarrow F(i), \exists! \text{ morphism} \\ X \rightarrow \lim F\end{aligned}$$

Conversely, given any morphism $\phi : X \rightarrow \lim F$, there is a unique corresponding family of morphisms $\{X \rightarrow F(i)\}$ given by the compositions $\{\theta_i \circ \phi\}$. In other words, there is a bijection between $\text{hom}_{\mathbf{C}}(X, \lim F)$ and the family of compatible morphisms from X to each $F(i)$.

The key observation here is that each family above, say $\{\xi_i\}$ can be interpreted as a natural transformation $\Delta_X \Rightarrow F$, and vice versa:

$$\begin{array}{ccc}\Delta_X(k) = X & \xrightarrow{\xi_k} & F(k) \\ \text{id}_X \downarrow & & \downarrow F(k \rightarrow \ell) \\ \Delta_X(\ell) = X & \xrightarrow{\xi_\ell} & F(\ell)\end{array}$$

Therefore, we have the following bijection:

$$\text{hom}_{\mathbf{C}}(X, \lim F) \cong \text{Nat}(\Delta_X, F)$$

This form suggests that we may have an adjoint pair $(\Delta, \lim)$ if we interpret $\lim$ as a functor $\mathbf{C}^{\mathbf{J}} \rightarrow \mathbf{C}$: $\lim$ takes a functor, say $F : \mathbf{J} \rightarrow \mathbf{C}$ as an input, and gives an object $\lim F$ in $\mathbf{C}$ as an output. Then, indeed, we have a natural isomorphism:

$$\mathrm{hom}_{\mathbf{C}}(-, \lim(-)) \xrightarrow{\sim} \mathrm{hom}_{\mathbf{C}^{\mathbf{J}}}(\Delta -, -)$$

Detailed verifications are omitted.

Dually, for colimits, we have the following bijection:

$$\mathrm{hom}_{\mathbf{C}}(\mathrm{colim} F, X) \cong \mathrm{Nat}(F, \Delta_X)$$

And hence, the natural isomorphism:

$$\mathrm{hom}_{\mathbf{C}}(\mathrm{colim}(-), -) \xrightarrow{\sim} \mathrm{hom}_{\mathbf{C}^{\mathbf{J}}}(-, \Delta_-)$$

Therefore, we have the following theorem:

Theorem 2.6.2. *$\lim$ (colim, resp) is the right (left, resp) adjoint of the diagonal functor Δ .*

3 Towards Abelian Categories

3.1 Additive Categories

Definition 3.1.1. A category $\mathbf{C}$ is said to be **additive** if it satisfies the following properties:

Ad1. For each $A, B \in \mathbf{C}$, $\text{hom}(A, B)$ is an abelian group, such that composition of morphisms distributes over addition. That is:

$$g \circ (f_1 + f_2) = g \circ f_1 + g \circ f_2 \quad (g_1 + g_2) \circ f = g_1 \circ f + g_2 \circ f$$

Ad2. $\mathbf{C}$ has a zero object, denoted as $\mathbf{0}$.

Ad3. It has products of two objects (a product $A \times B$ for any pair of objects), and hence by induction, products of any finite number of objects.

Remark. As a consequence of this definition, in additive categories, finite direct products and finite direct sums coincide. The symbol $\oplus$ is used for this notion.

Remark. In an additive category, the morphisms are called **homomorphisms**, and $\text{hom}(-, -)$ is denoted as $\text{Hom}(-, -)$, as we actually did in the previous sections. A functor F between additive categories preserving the additive structures of Hom is called an **additive functor**. Namely,

$$F(f + g) = F(f) + F(g)$$

Remark. In an additive category, the **zero map** between objects X and Y , denoted as $0 : X \longrightarrow Y$, is just the identity of the abelian group $\text{Hom}(X, Y)$. Whereas in general, a zero map only makes sense when the category has a zero object, and is then defined as the composition:

$$X \xrightarrow{!} \mathbf{0} \xrightarrow{!} Y$$

Example 3.1.2. Two examples of additive categories:

1. The category of Banach spaces over a field $\mathbb{K} = \mathbb{C}$ or $\mathbb{R}$, denoted as $\mathbf{Ban}_{\mathbb{K}}$.
2. The category of Free modules over a ring A , denoted as $\mathbf{FMod}_A$.

3.2 Kernels and Cokernels

Definition 3.2.1. Let $\mathbf{C}$ be a category with a $\mathbf{0}$ object. The **kernel** of a morphism $f : B \longrightarrow C$ is a morphism $\iota : \ker f \longrightarrow B$ (together with the object $\ker f$) such that $f \circ \iota = 0$. It is universal

w.r.t. this property. That is:

$$\begin{array}{ccccc}
 Z & & & & \\
 \downarrow \exists! & \searrow & & \searrow & \\
 \ker f & \xrightarrow{\iota} & B & \xrightarrow{f} & C \\
 & \nearrow & & \nearrow & \\
 & & & & 0
 \end{array}$$

Hence, it is unique up to unique isomorphism. Another interpretation: The kernel of $f : B \rightarrow C$, is the limit of the diagram:

$$\begin{array}{ccc}
 & \mathbf{0} & \\
 & \downarrow & \\
 B & \xrightarrow{f} & C
 \end{array}$$

Definition 3.2.2. Dually, the **cokernel** of a morphism $f : B \rightarrow C$ is a morphism $\pi : C \rightarrow \text{coker } f$ (together with the object $\text{coker } f$) such that $\pi \circ f = 0$. It is universal w.r.t. this property. That is:

$$\begin{array}{ccccc}
 & & & & Z \\
 & & & \nearrow & \uparrow \exists! \\
 B & \xrightarrow{f} & C & \xrightarrow{\pi} & \text{coker } f \\
 & \searrow & & \searrow & \\
 & & & & 0
 \end{array}$$

Hence it is unique up to unique isomorphism. Another interpretation: The cokernel of $f : B \rightarrow C$, is the colimit of the diagram:

$$\begin{array}{ccc}
 B & \xrightarrow{f} & C \\
 \downarrow & & \\
 \mathbf{0} & &
 \end{array}$$

Remark. In additive categories, kernels are monic, and cokernels are epic.

Proof. It suffices to show the case in kernels; the case in cokernels follows symmetrically. Let ι , together with $\ker f$, be the kernel of $f : B \rightarrow C$. Suppose $\iota \circ g = \iota \circ h$ where $g, h \in \text{hom}(X, \ker f)$. Then:

$$\begin{array}{ccccc}
 X & & & & \\
 \downarrow g & \searrow \iota \circ g & & \searrow & \\
 \ker f & \xrightarrow{\iota} & B & \xrightarrow{f} & C \\
 \uparrow h & \nearrow \iota \circ h & & \nearrow & \\
 X & & & & 0
 \end{array}$$

So $(\iota \circ g) \circ f = (\iota \circ h) \circ f = 0$. By the universal property of $\ker f$, $\exists!$ morphism from X to $\ker f$, forcing g and h to be the same. $\square$

3.3 Abelian Categories

Definition 3.3.1. An **Abelian category** is an additive category satisfying three additional properties:

Ab1. Every morphism has a kernel and a cokernel.

Ab2. Every monomorphism is the kernel of its cokernel.

Ab3. Every epimorphism is the cokernel of its kernel.

Remark. By the previous remark, in an abelian category, monic = “is a kernel” and epic = “is a cokernel.” In general, in an abelian category,

$$\text{monic} + \text{epic} \implies \text{isomorphism}$$

In particular, in $\mathbf{Mod}_A$,

$$\text{monic} = \text{injective} \quad \text{epic} = \text{surjective}$$

Definition 3.3.2. Now we can define the **image** and the **coimage** of an arbitrary morphism, say $f : A \longrightarrow B$:

$$\text{im } f := \ker(\text{coker } f) \quad \text{coim } f := \text{coker}(\ker f)$$

Remark. The two examples in section 3.1.2 are not an abelian category.

1. In $\mathbf{Ban}_{\mathbb{K}}$, the image of a bounded linear operator $T : X \longrightarrow Y$ may not be closed; hence, the cokernel $Y/\text{im } T$ may not be complete and may not be in $\mathbf{Ban}_{\mathbb{K}}$.
2. In $\mathbf{FMod}_A$, the kernel of a morphism may not be free.

Theorem 3.3.3. *In an abelian category, given a morphism $f : A \longrightarrow B$, there is a commutative diagram*

$$\begin{array}{ccccccc} \ker f & \xrightarrow{\iota} & A & \xrightarrow{f} & B & \xrightarrow{\pi} & \text{coker } f \\ \downarrow & & \downarrow & & \uparrow & & \uparrow \\ \mathbf{0} & \longrightarrow & \text{coim } f & \xlongequal{\quad} & \text{im } f & \longrightarrow & \mathbf{0} \end{array}$$

where the left and right squares are the pullback-pushout squares, and the double line is a unique isomorphism. As a result, any morphism $f : A \longrightarrow B$ can be factored uniquely as:

$$A \xrightarrow{\text{epic}} \text{coim } f \xrightarrow{\cong} \text{im } f \xrightarrow{\text{monic}} B$$

In fact, we may treat every abelian category as $\mathbf{Mod}_A$ where A is a ring (not necessarily commutative!) in some sense, thanks to the following theorem:

Theorem 3.3.4 (Freyd-Mitchell Embedding). *Let $\mathbf{C}$ be a locally small abelian category. Then $\exists$ a ring A and an exact, fully faithful functor $F : \mathbf{C} \longrightarrow \mathbf{Mod}_A$, which therefore embeds $\mathbf{C}$ as a full subcategory of $\mathbf{Mod}_A$.*

3.4 Some Homological Algebra, FHHF, RAPL, LAPC, etc.

From now on, we work in an abelian category.

Definition 3.4.1. We say a sequence

$$\cdots \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow \cdots$$

is **complex at** B iff $g \circ f = 0$, and is **exact at** B iff $\ker g = \operatorname{im} f$. A sequence is a **complex** (resp. **exact**) iff it is complex (resp. exact) at each internal term. If the above sequence is complex at B , then its **homology** at B is $\ker g / \operatorname{im} f$. We say the elements of $\ker g$ are the **cycles**, and the elements in $\operatorname{im} f$ are the **boundaries**. If the complex is indexed in decreasing order, the indices are often written as subscripts, and H_i is the homology at $A_{i+1} \longrightarrow A_i \longrightarrow A_{i-1}$. If the complex is indexed in increasing order, the indices are often written as superscripts, and the homology H^i at $A^{i-1} \longrightarrow A^i \longrightarrow A^{i+1}$ is called **cohomology**.

An exact sequence

$$A^\bullet : \quad \cdots \longrightarrow A^{i-1} \xrightarrow{f^{i-1}} A^i \xrightarrow{f^i} A^{i+1} \xrightarrow{f^{i+1}} \cdots$$

can be factored into short exact sequences

$$0 \longrightarrow \ker f^i \longrightarrow A^i \longrightarrow \ker f^{i+1} \longrightarrow 0$$

In particular, we have the following useful fact:

Proposition 3.4.2. *Let*

$$0 \longrightarrow A \xrightarrow{g} B \xrightarrow{f} C$$

Be an exact sequence, then $A \cong \ker f$.

Proof. By the universal property of kernels, g factors through $\ker f$ uniquely, say $g = \iota \circ \phi$.

$$\begin{array}{ccccc}
 & & \operatorname{im} g & & \\
 & \uparrow \psi & \nearrow \iota & & \\
 0 & \longrightarrow & A & \xrightarrow{g} & B \xrightarrow{f} C \\
 & \downarrow \exists! \text{ say } \phi & \nwarrow \iota & & \\
 & & \ker f & &
 \end{array}$$

Since g is monic, ϕ has to be monic. Also, note that g can be factored as $A \twoheadrightarrow \operatorname{im} g \hookrightarrow B$. But since $\operatorname{im} g = \ker f$, $\psi = \phi$, hence ϕ is monic and epic. Since in abelian categories, monic + epic = iso, $\ker g \cong A$. $\square$

More generally, if the $A^\bullet$ above is assumed only to be a complex, then it can be factored into short exact sequences:

$$0 \longrightarrow \ker f^i \longrightarrow A^i \longrightarrow \operatorname{im} f^i \longrightarrow 0$$

$$0 \longrightarrow \operatorname{im} f^{i-1} \longrightarrow \ker f^i \longrightarrow H^i(A^\bullet) \longrightarrow 0$$

Definition 3.4.3. Suppose

$$0 \xrightarrow{d^0} A^1 \xrightarrow{d^1} \dots \xrightarrow{d^{n-1}} A^n \xrightarrow{d^n} 0$$

is a complex of finite-dimensional k -vector spaces, denoted as $A^\bullet$. Define

$$h^i(A^\bullet) := \dim H^i(A^\bullet)$$

We claim that

$$\sum (-1)^i \dim A^i = \sum (-1)^i h^i(A^\bullet)$$

In particular, if $A^\bullet$ is exact, then

$$\sum (-1)^i \dim A^i = 0$$

Proof. Straightforward. Note that $\dim \operatorname{im} d^i = \dim A^i - \dim \ker d^i$. □

Definition 3.4.4. Let $\mathbf{A}$ be an abelian category. Define the category $\mathbf{Ch}_{\mathbf{A}}$ of complexes as follows: The objects are infinite complexes:

$$A^\bullet : \quad \dots \longrightarrow A^{i-1} \xrightarrow{f^{i-1}} A^i \xrightarrow{f^i} A^{i+1} \xrightarrow{f^{i+1}} \dots$$

A morphism $\Phi : A^\bullet \longrightarrow B^\bullet$ is a commuting diagram:

$$\begin{array}{ccccccc} \dots & \longrightarrow & A^{i-1} & \xrightarrow{f^{i-1}} & A^i & \xrightarrow{f^i} & A^{i+1} \xrightarrow{f^{i+1}} \dots \\ & & \downarrow \varphi^{i-1} & & \downarrow \varphi^i & & \downarrow \varphi^{i+1} \\ \dots & \longrightarrow & B^{i-1} & \xrightarrow{g^{i-1}} & B^i & \xrightarrow{g^i} & B^{i+1} \xrightarrow{g^{i+1}} \dots \end{array}$$

We claim that $\mathbf{Ch}_{\mathbf{A}}$ is an abelian category. To verify this, we may take $\mathbf{A}$ as $\mathbf{Mod}_R$ for some ring R , thanks to Freyd-Mitchell 3.3.4. Basically, everything can be carried degreewise; we omit the details.

Theorem 3.4.5 (Homology as a functor). *We inherit the diagram above.*

A morphism $\Phi : A^\bullet \longrightarrow B^\bullet$ in $\mathbf{Ch}_{\mathbf{A}}$ induces a family of morphisms $\phi^i : H^i(A^\bullet) \longrightarrow H^i(B^\bullet)$ in $\mathbf{A}$ for each i . Furthermore, H^i can be viewed as a covariant functor $\mathbf{Ch}_{\mathbf{A}} \longrightarrow \mathbf{A}$.

Proof. Again, we may set $\mathbf{A}$ as $\mathbf{Mod}_R$. Fix an index i , we shall specify the map ϕ^i first. Define:

$$\phi^i : \quad H^i(A^\bullet) \longrightarrow H^i(B^\bullet)$$

$$\bar{c} := c + \operatorname{im} f^{i-1} \longmapsto \varphi^i(c) + \operatorname{im} g^{i-1}$$

(i). We need to show the well-definedness. First, $\varphi^i(c) \in \ker g^i$ by the commutativity of the second square. Now, suppose $\overline{c_1} = \overline{c_2}$, then $c_1 - c_2 \in \operatorname{im} f^{i-1} = \ker f^i$. Again, by the commutativity of the second square, $g^i(\varphi^i(c_1 - c_2)) = 0$, so $\varphi^i(c_1) - \varphi^i(c_2) \in \ker g^i = \operatorname{im} g^{i-1}$. Hence $\phi^i(\overline{c_1}) = \phi^i(\overline{c_2})$, as desired. (ii). Then, we need to show that ϕ^i is indeed a morphism, which is just a straightforward verification. (iii). Finally, the functorial property can also be easily read off from the definition. $\square$

Theorem 3.4.6 (Short exact sequence of complexes induces long exact sequence of homology). *Suppose we have a short exact sequence of complexes:*

$$0^\bullet \longrightarrow A^\bullet \longrightarrow B^\bullet \longrightarrow C^\bullet \longrightarrow 0^\bullet$$

This induces a long exact sequence in cohomology:

$$\begin{array}{ccccccc} & & & & \cdots & \longrightarrow & H^{i-1}(C^\bullet) \\ & & & & & \swarrow \text{dashed} & \\ H^i(A^\bullet) & \xrightarrow{\quad} & H^i(B^\bullet) & \longrightarrow & H^i(C^\bullet) & & \\ & & & & \nwarrow \text{dashed} & & \\ H^{i+1}(A^\bullet) & \xrightarrow{\quad} & \cdots & & & & \end{array}$$

where the horizontal arrows are just the induced morphisms from our previous theorem, and the dashed arrows are called the **connecting homomorphism**, which need to be defined properly; we postpone the further discussion of this.

Theorem 3.4.7 (FHHF). *Let $F : \mathbf{A} \longrightarrow \mathbf{B}$ be a covariant functor of abelian categories, and $C^\bullet$ a complex in $\mathbf{A}$. Then*

(a). *F right exact yields a natural morphism $FH^\bullet \longrightarrow H^\bullet F$ in $\mathbf{B}$.*

More precisely, for each i , the left side is F applied to the cohomology at piece i of $C^\bullet$, while the right side is the cohomology at piece i of $FC^\bullet$.

(b). *F left exact yields a natural morphism $FH^\bullet \longleftarrow H^\bullet F$ in $\mathbf{B}$.*

(c). *F exact yields $FH^\bullet \xrightarrow{\sim} H^\bullet F$.*

The morphisms of (a) and (b) are inverses and thus isomorphisms.

Proof. Let $C^\bullet : \cdots \longrightarrow C^{i-1} \xrightarrow{d^{i-1}} C^i \xrightarrow{d^i} C^{i+1} \longrightarrow \cdots$

Note that we have the following two exact sequences:

$$0 \longrightarrow \operatorname{im} d^i \longrightarrow C^{i+1} \longrightarrow \operatorname{coker} d^i \longrightarrow 0 \quad \dagger$$

$$0 \longrightarrow H^i(C^\bullet) \longrightarrow \operatorname{coker} d^{i-1} \longrightarrow \operatorname{im} d^i \longrightarrow 0 \quad \ddagger$$

Note that both $\dagger$ and $\ddagger$ are of the form:

$$\mathbf{0} \longrightarrow \clubsuit \xrightarrow{\iota} \spadesuit \xrightarrow{\pi} \spadesuit / \clubsuit \longrightarrow \mathbf{0}$$

where ι is the embedding, π is the projection. In fact

$$\mathbf{0} \longrightarrow \ker \pi \xrightarrow{\iota} \spadesuit \xrightarrow{\pi} \text{coker } \iota \longrightarrow \mathbf{0}$$

(a). Note that $C^i \xrightarrow{d^i} C^{i+1} \longrightarrow \text{coker } d^i \longrightarrow \mathbf{0}$ is exact. Applying F to it, the sequence

$$FC^i \xrightarrow{Fd^i} FC^{i+1} \longrightarrow F \text{coker } d^i \longrightarrow \mathbf{0}$$

is exact. Hence, $\text{coker } Fd^i \cong F \text{coker } d^i$. And the following sequence

$$\text{im } Fd^i \xrightarrow{\text{embedding}} FC^{i+1} \longrightarrow F \text{coker } d^i \longrightarrow \mathbf{0}$$

is exact. Applying F to $\dagger$, the sequence

$$F \text{im } d^i \xrightarrow{\mu, \text{ say}} FC^{i+1} \longrightarrow F \text{coker } d^i \longrightarrow \mathbf{0}$$

is exact. This gives an epimorphism:

$$F \text{im } d^i \twoheadrightarrow \text{im } Fd^i$$

Then, applying F to $\ddagger$, the sequence

$$FH^i(C^\bullet) \longrightarrow F \text{coker } d^{i-1} \longrightarrow F \text{im } d^i \longrightarrow \mathbf{0}$$

is exact. Apply $\ddagger$ to $FC^\bullet$, the following sequence

$$\mathbf{0} \longrightarrow H^i(FC^\bullet) \longrightarrow \text{coker } Fd^{i-1} \longrightarrow \text{im } Fd^i \longrightarrow \mathbf{0}$$

is exact. Hence, we have the following diagram:

$$\begin{array}{ccccccc} FH^i(C^\bullet) & \xrightarrow{\quad \quad} & F \text{coker } d^{i-1} & \xrightarrow{\quad \quad} & F \text{im } d^i & \longrightarrow & \mathbf{0} \\ & \searrow \text{blue} & \parallel \text{blue} & & \downarrow \text{purple} & & \\ H^i(FC^\bullet) & \longrightarrow & \text{coker } Fd^{i-1} & \xrightarrow{\theta} & \text{im } Fd^i & \longrightarrow & \mathbf{0} \end{array}$$

where the two rows are exact.

The key observation is that the square above commutes. (Why?)

Consequently, the composition:

$$* \xrightarrow{\text{blue}} * \xrightarrow{\cong} * \xrightarrow{\theta} *$$

is a zero map. Hence, $FH^i(C^\bullet)$ is in the kernel of θ , which is exactly $H^i(FC^\bullet)$.

(b). Dually, we would eventually obtain a commuting diagram:

$$\begin{array}{ccccccc}
\mathbf{0} & \longrightarrow & F \operatorname{im} d^{i-1} & \longrightarrow & F \operatorname{ker} d^i & \longrightarrow & FH^i(C^\bullet) \\
& & \uparrow & & \parallel & & \\
\mathbf{0} & \longrightarrow & \operatorname{im} F d^{i-1} & \longrightarrow & \operatorname{ker} F d^i & \longrightarrow & H^i(FC^\bullet)
\end{array}$$

Then use a dual argument to yield the result. We omit the details.

(c). TBD.....

Part (c) is actually saying:

An exact functor commutes with taking cohomology.

□

The rest of this note will be boiled down to the following statements:

Limits commute with limits and right adjoints.

In particular, since in an abelian category, kernels are limits, both right adjoints and limits are left-exact.

And dually,

Colimits commute with colimits and left adjoints.

In particular, since in an abelian category, cokernels are colimits, both left adjoints and limits are right-exact.

Theorem 3.4.8 (Kernels commute with limits). *Let $\mathbf{A}$ be an abelian category, and $a, b : \mathbf{I} \rightarrow \mathbf{A}$ are two diagrams in $\mathbf{A}$ indexed by $\mathbf{I}$. Let $A_i := a(i)$ and $B_i := b(i)$ be the objects in those two diagrams. Let $h_i : A_i \rightarrow B_i$ be morphisms commuting with the morphisms in the diagram. (That is, h is a natural transformation $a \Rightarrow b$.) Then $\{\operatorname{ker} h_i\}$ forms another diagram in $\mathbf{A}$ indexed by $\mathbf{I}$. There is a canonical isomorphism*

$$\varprojlim_{\mathbf{I}} \operatorname{ker} h_i \cong \operatorname{ker} \left(\varprojlim_{\mathbf{I}} A_i \longrightarrow \varprojlim_{\mathbf{I}} B_i \right)$$

assuming the limits exist.

Proof. Step 1. We need to specify the morphisms between the kernels. Since h is a natural transformation, the right square of the following diagram commutes:

$$\begin{array}{ccccc}
 \ker h_j & \xrightarrow{\quad \quad} & A_j & \xrightarrow{h_j} & B_j \\
 \downarrow & & \downarrow & & \downarrow \\
 \ker h_k & \xrightarrow{\quad \quad} & A_k & \xrightarrow{h_k} & B_k
 \end{array}$$

Consequently, the composition:

$$* \xrightarrow{\quad \quad} * \xrightarrow{\quad \quad} * \xrightarrow{h_k} *$$

is a zero map. Hence, by the universal property of $\ker h_k$, we have a unique morphism, say $\xi_{jk} : \ker h_j \rightarrow \ker h_k$ s.t. $\iota_k \circ \xi_{jk} = \mu_{jk} \circ \iota_j$, and this applies to all the morphisms in $\mathbf{I}$. Therefore, for any morphism $j \rightarrow k$ in $\mathbf{I}$, the following induced diagram commutes:

$$\begin{array}{ccccc}
 \ker h_j & \xrightarrow{\iota_j} & A_j & \xrightarrow{h_j} & B_j \\
 \downarrow \xi_{jk} & & \downarrow \mu_{jk} & & \downarrow \nu_{jk} \\
 \ker h_k & \xrightarrow{\iota_k} & A_k & \xrightarrow{h_k} & B_k
 \end{array}$$

Step 2. We need to specify the morphism $\varprojlim A_i \rightarrow \varprojlim B_i$. Indeed, since $\{h_i \circ \alpha_i\}$ forms a family of compatible morphisms towards $\{B_i\}$, by the universal property of $\varprojlim B_i$, $\exists!$ morphism, say

$$\Theta : \varprojlim A_i \rightarrow \varprojlim B_i$$

s.t. $h_i \circ \alpha_i = \beta_i \circ \Theta$, for each $i \in \mathbf{I}$.

Step 3. By our assumption, we therefore have an object $K := \varprojlim \ker h_i$, together with a family of morphisms $\{\gamma_i : K \rightarrow \ker h_i\}$. Then, similar to step 2, we have a canonical morphism, say

$$\Delta : K \rightarrow \varprojlim A_i$$

s.t. $\iota_i \circ \gamma_i = \alpha_i \circ \Delta$, for each $i \in \mathbf{I}$.

Summarize our previous arguments, the following diagram commutes:

$$\begin{array}{ccccc}
 K & \xrightarrow{\Delta} & \varprojlim A_i & \xrightarrow{\Theta} & \varprojlim B_i \\
 \swarrow \gamma_j & & \swarrow \alpha_j & & \swarrow \beta_j \\
 \ker h_j & \xrightarrow{\iota_j} & A_j & \xrightarrow{h_j} & B_j \\
 \downarrow \xi_{jk} & & \downarrow \mu_{jk} & & \downarrow \nu_{jk} \\
 \ker h_k & \xrightarrow{\quad \quad} & A_k & \xrightarrow{\quad \quad} & B_k
 \end{array}$$

Step 4. Now, it remains to show that the object K , [is] indeed $\ker \Theta$. That is:

We want to show: $\exists!$ isomorphism $K \rightarrow \ker \Theta$.

We first state a lemma:

Lemma 3.4.9. Let $f, g : X \longrightarrow \varprojlim B_i$. If for any $i \in \mathbf{I}$ we always have

$$\beta_i \circ f = \beta_i \circ g$$

Then $f = g$. We say the family $\{\beta_i\}_{i \in \mathbf{I}}$ is **jointly monic**.

Proof of this lemma. This can be obtained directly from the universal property of $\varprojlim B_i$. $\square$

Now, Note that

$$0 = \beta_i \circ (\Theta \circ \Delta) = \beta_i \circ 0 \quad \forall i \in \mathbf{I}$$

Hence, by our lemma, $\Theta \circ \Delta$ is a zero map. Consequently, by the universal property of $\ker \Theta$,

$$\exists! \text{ morphism, say } \Phi : K \longrightarrow \ker \Theta.$$

s.t. $\iota \circ \Phi = \Delta$ where ι is the kernel of Θ .

Note that for each $i \in \mathbf{I}$, thanks to the commutativity of the above diagram, the composition

$$\ker \Theta \xrightarrow{\iota} \varprojlim A_i \xrightarrow{\alpha_i} A_i \xrightarrow{h_i} B_i$$

is a zero map. Hence, by the universal property of $\ker h_i$, the composition $\alpha_i \circ \iota$ induces a morphism, say

$$\overline{\gamma}_i : \ker \Theta \longrightarrow \ker h_i$$

s.t. $\iota_i \circ \overline{\gamma}_i = \alpha_i \circ \iota$

Now, in order to define the converse morphism $\Psi : \ker \Theta \longrightarrow K$ using the universal property of $\varprojlim \ker h_i$, we need to show that the $\{\overline{\gamma}_i\}$ are compatible, namely, $\xi_{jk} \circ \overline{\gamma}_j = \overline{\gamma}_k$ for any morphism $j \longrightarrow k$ in $\mathbf{I}$. To do this, note that for any $j \longrightarrow k$, we have $\mu_{jk} \circ \alpha_j \circ \iota = \alpha_k \circ \iota$, so $\{\alpha_i \circ \iota\}$ are compatible. But since as we just showed, $\iota_i \circ \overline{\gamma}_i = \alpha_i \circ \iota$, we obtain that $\{\iota_i \circ \overline{\gamma}_i\}$ are compatible. Namely,

$$\mu_{ik} \circ \iota_j \circ \overline{\gamma}_j = \iota_k \circ \overline{\gamma}_k$$

for each $j \longrightarrow k$. But note that $\mu_{ik} \circ \iota_j = \iota_k \circ \xi_{jk}$, so $\iota_k \circ \xi_{jk} \circ \overline{\gamma}_j = \iota_k \circ \overline{\gamma}_k$. That is, in the following diagram, the cyan and the violet compositions are the same morphism:

$$\begin{array}{ccc}
 & \ker \Theta & \\
 \overline{\gamma}_j \swarrow & & \searrow \overline{\gamma}_k \\
 \ker h_j & \xrightarrow{\xi_{jk}} & \ker h_k \\
 \downarrow \iota_j & & \downarrow \iota_k \\
 A_j & \xrightarrow{\mu_{jk}} & A_k
 \end{array}$$

But since ι_k is monic, we conclude that

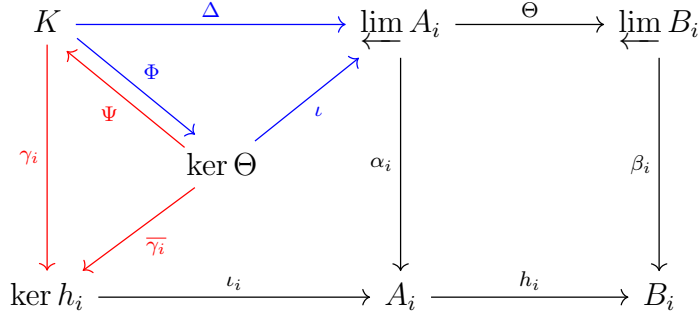
$$\xi_{jk} \circ \overline{\gamma}_j = \overline{\gamma}_k$$

for each $j \longrightarrow k$, as desired. Therefore,

$$\exists! \text{ morphism } \Psi : \ker \Theta \longrightarrow K.$$

s.t. $\gamma_i \circ \Psi = \overline{\gamma_i}$ for each i in $\mathbf{I}$.

Finally, it remains to show that $\Psi \circ \Phi = \text{id}$ and $\Phi \circ \Psi = \text{id}$. Keep in mind the following diagram. Note that the red and blue triangles are commuting diagrams.



To show $\Phi \circ \Psi = \text{id}$, note that

$$\begin{aligned}
& \Phi \circ \Psi = \text{id} \\
\iff & \iota \circ \Phi \circ \Psi = \iota \circ \text{id} && \text{since } \iota \text{ is monic} \\
\iff & \Delta \circ \Psi = \iota \\
\iff & \alpha_i \circ \Delta \circ \Psi = \alpha_i \circ \iota && \text{for each } i \in \mathbf{I}, \text{ since } \{\alpha_i\} \text{ is jointly monic} \\
\iff & \iota_i \circ \gamma_i \circ \Psi = \iota_i \circ \overline{\gamma_i} && \text{for each } i \in \mathbf{I} \\
\iff & \gamma_i \circ \Psi = \overline{\gamma_i} && \text{for each } i \in \mathbf{I}
\end{aligned}$$

which is true by the commutativity of the red triangle.

To show $\Psi \circ \Phi = \text{id}$, note that

$$\begin{aligned}
& \Psi \circ \Phi = \text{id} \\
\iff & \gamma_i \circ \Psi \circ \Phi = \gamma_i \circ \text{id} && \text{for each } i \in \mathbf{I}, \text{ since } \{\gamma_i\} \text{ is jointly monic} \\
\iff & \overline{\gamma_i} \circ \Phi = \gamma_i \\
\iff & \iota_i \circ \overline{\gamma_i} \circ \Phi = \iota_i \circ \gamma_i && \text{since } \iota_i \text{ is monic, for all } i \in \mathbf{I} \\
\iff & \alpha_i \circ \iota \circ \Phi = \alpha_i \circ \Delta \\
\iff & \iota \circ \Phi = \Delta
\end{aligned}$$

which is true by the commutativity of the blue triangle.

Remark. The final step actually shows that the above diagram commutes.

□

Since kernels are limits, the above theorem is just a corollary of the following theorem:

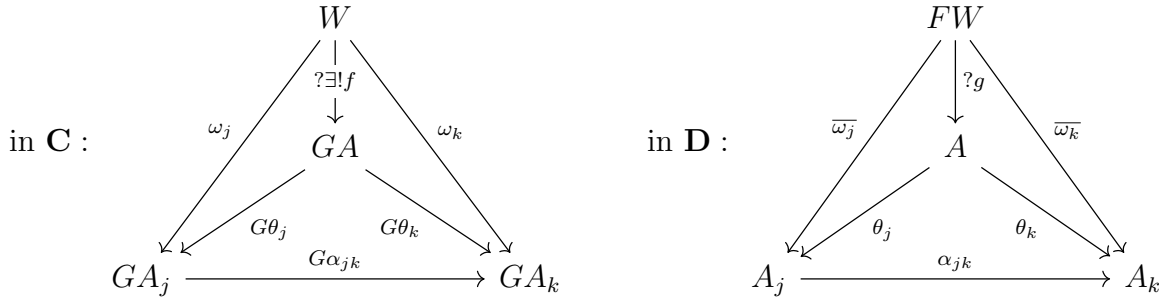
Theorem 3.4.10 (limits commute with limits). *Let $\mathbf{I}, \mathbf{J}$ be small categories, $\mathbf{C}$ be an arbitrary category. Let $F : \mathbf{I} \times \mathbf{J} \rightarrow \mathbf{C}$ be a functor. (see definition 2.3.1 for the definition of product categories.) Assume all limits exist. Then there is a canonical isomorphism*

$$\varprojlim_{\mathbf{I}} \varprojlim_{\mathbf{J}} F(i, j) \cong \varprojlim_{\mathbf{J}} \varprojlim_{\mathbf{I}} F(i, j)$$

Theorem 3.4.11 (right adjoints commute with limits). *Let (F, G) be a pair of adjoint functors, where $F : \mathbf{C} \rightarrow \mathbf{D}$ and $G : \mathbf{D} \rightarrow \mathbf{C}$. Let $A := \varprojlim A_i$ be the limit in $\mathbf{D}$ of a diagram, then GA , with the corresponding maps $G(\theta_i) : GA \rightarrow GA_i$, where $\theta_i : A \rightarrow A_i$, is the limit in $\mathbf{C}$, w.r.t. the same index category. In other words,*

$$GA = \varprojlim GA_i$$

Proof. We need to show that GA satisfies the universal property of $\varprojlim GA_i$. Suppose an object W , together with morphisms $\{\omega_i : W \rightarrow GA_i\}$ in $\mathbf{C}$, is also compatible with $\{\alpha_{jk} : A_j \rightarrow A_k\}$. We must show that $\exists!$ morphism, say $f : W \rightarrow GA$ s.t. $G(\theta_i) \circ f = \omega_i$ for each i . Since (F, G) is an adjoint pair, this is equivalent to showing: $\exists!$ morphism, say $g : FW \rightarrow A$ extending the $\{\theta_i\}$, but this is exactly the universal property of the limit. To be more precise,



We know that there is a natural isomorphism, say

$$\Phi : \text{hom}_{\mathbf{C}}(-, G(-)) \xrightarrow{\sim} \text{hom}_{\mathbf{D}}(F(-), -)$$

We need to show that $\{\bar{\omega}_i : FW \rightarrow A_i\}$ where $\bar{\omega}_i := \Phi_{W, A_i}(\omega_i)$ are compatible with $\{\alpha\}$. Note that the following diagram

$$\begin{array}{ccc}
\text{hom}_{\mathbf{C}}(W, GA_j) & \xrightarrow{\Phi_{W, A_j}} & \text{hom}_{\mathbf{D}}(FW, A_j) \\
\downarrow G\alpha_{jk} \circ - & & \downarrow \alpha_{jk} \circ - \\
& \begin{array}{ccc} \omega_j & \xrightarrow{\quad} & \bar{\omega}_j \\ \downarrow & & \downarrow \\ G\alpha_{jk} \circ \omega_j & \xrightarrow{\quad} & \overline{G\alpha_{jk} \circ \omega_j} \end{array} & \\
\text{hom}_{\mathbf{C}}(W, GA_k) & \xrightarrow{\Phi_{W, A_k}} & \text{hom}_{\mathbf{D}}(FW, A_k)
\end{array}$$

commutes. Hence

$$\bar{\omega}_k = \overline{G\alpha_{jk} \circ \omega_j} = \alpha_{jk} \circ \bar{\omega}_j$$

as desired. $\square$

Remark. In [2], they call this fact and its dual:

Right adjoints preserve limits, left adjoints preserve colimits.

and denote them as RAPL and LAPC, resp.

Remark. By Theorem 2.6.2, we know that limits are right adjoints. Therefore, since right adjoints commute with limits, we have Theorem 3.4.10: limits commute with limits for free. Similarly, we obtain that colimits commute with colimits.

Corollary 3.4.12. *If F and G are additive functors between abelian categories, and (F, G) is an adjoint pair, then (as kernels are limits and cokernels are colimits) G is left-exact, and F is right-exact.*

Proof. Note that right adjoints commute with limits, and kernels are limits, so right adjoints commute with kernels. Suppose the following sequence

$$0 \longrightarrow A \xrightarrow{g} B \xrightarrow{f} C$$

is exact. Then by Proposition 3.4, this is equivalent to saying

$$0 \longrightarrow \ker f \xrightarrow{\iota} B \xrightarrow{f} C$$

is exact. Applying G , we obtain:

$$0 \longrightarrow G\ker f \xrightarrow{G\iota} GB \xrightarrow{Gf} GC$$

Since G commutes with kernels, there is a canonical isomorphism $G\ker f \cong \ker Gf$. Under this identification, $G\iota$ is the kernel of Gf . Hence,

$$0 \longrightarrow \ker Gf \xrightarrow{G\iota} GB \xrightarrow{Gf} GC$$

is exact. So G is left-exact. □

Corollary 3.4.13. *Limits are left-exact, and dually, colimits are right-exact.*

Proof. By the previous corollary, we know that right adjoints are left-exact; since limits are right adjoints, they are left-exact. □

To summarize the above discussions, we have the following:

$$\begin{array}{ccccc}
 \text{RA comm.w/ limits (RAPL)} & \implies & \text{RA comm.w/ kernels} & \implies & \text{RA are left-exact} \\
 \Downarrow & & \Downarrow & & \Downarrow \\
 \text{limits comm.w/ limits} & \implies & \text{limits comm.w/ kernels} & \implies & \text{limits are left-exact}
 \end{array}$$

And dually,

$$\begin{array}{ccccc}
\text{LA comm.w/ colimits (LAPC)} & \Longrightarrow & \text{LA comm.w/ cokernels} & \Longrightarrow & \text{LA are right-exact} \\
\Downarrow & & \Downarrow & & \Downarrow \\
\text{colimits comm.w/ colimits} & \Longrightarrow & \text{colimits comm.w/ cokernels} & \Longrightarrow & \text{colimits are right-exact}
\end{array}$$

Theorem 3.4.14. *In $\mathbf{Mod}_A$, colimits over filtered index categories are exact.*

Proof. Right-exactness follows from the above discussion, so it suffices to show the left-exactness. To show this, let h be a natural transformation where $h_i : M_i \rightarrow N_i, i \in \mathbf{I}$. It suffices to show that

$$\varinjlim_{\mathbf{I}} \ker h_i \cong \ker \left(\varinjlim_{\mathbf{I}} M_i \rightarrow \varinjlim_{\mathbf{I}} N_i \right)$$

Because if this is true, then filtered colimits commute with kernels, hence are left-exact.

Recall the construction of filtered colimits:

$$\varinjlim_{\mathbf{I}} M_i := \bigsqcup_{i \in \mathbf{I}} M_i / \sim$$

where

$$m_i \sim m_j \iff \exists k, \text{ w/ arrows } i \rightarrow k, j \rightarrow k \text{ they are identical in } M_k$$

Here, the relation $\sim$ is guaranteed to be an equivalent relation thanks to the filtered-ness of $\mathbf{I}$.

Now, similar to the proof of Theorem 3.4.8, we have an induced morphism:

$$\Delta : \varinjlim_{\mathbf{I}} \ker h_i \rightarrow \varinjlim_{\mathbf{I}} M_i$$

By a similar reasoning in the proof of Theorem 3.4.8, we know $\Theta \circ \Delta = 0$. So $\text{im } \Delta \subseteq \ker \Theta$.

$$\begin{array}{ccccc}
\ker h_i & \hookrightarrow & M_i & \xrightarrow{h_i} & N_i \\
\downarrow & & \downarrow & & \downarrow \\
\varinjlim_{\mathbf{I}} \ker h_i & \xrightarrow{\Delta} & \varinjlim_{\mathbf{I}} M_i & \xrightarrow{\Theta} & \varinjlim_{\mathbf{I}} N_i
\end{array}$$

So we have the restriction map

$$\overline{\Delta} : \varinjlim_{\mathbf{I}} \ker h_i \rightarrow \ker \Theta$$

Our goal is to show that $\overline{\Delta}$ is a bijection.

Surjectivity: let $[m_i] \in \ker \Theta$, then $[h_i(m_i)] = 0$ in $\varinjlim_{\mathbf{I}} N_i$ thanks to the commuting diagram above. Hence, $\exists$ an arrow $i \rightarrow j$ for some j s.t. the image of $h_i(m_i)$ in N_j is 0.

$$\begin{array}{ccc} M_i & \xrightarrow{h_i} & N_i \\ \downarrow & & \downarrow \\ M_j & \xrightarrow{h_j} & N_j \end{array}$$

Let m_j be the image of m_i in M_j . Then, thanks to the commuting diagram above, $h_j(m_j) = 0$. That is, $m_j \in \ker h_j$, hence $[m_j] \in \varinjlim_{\mathbf{I}} \ker h_i$. But $[m_i] = [m_j]$; hence, $[m_i]$ is from $\varinjlim_{\mathbf{I}} \ker h_i$, as desired.

Injectivity: Let $[x_i] \in \ker \overline{\Delta}$, where $x_i \in \ker h_i$ (embedded in M_i). This means the image of x_i is zero in some later M_j . But $\ker h_i \rightarrow \ker h_j$ is the restriction induced by $M_i \rightarrow M_j$, so the image of x_i in $\ker h_j$ is also zero. Hence, $[x_i] = 0$, as desired. $\square$

Corollary 3.4.15. *filtered colimits commute with homology in $\mathbf{Mod}_A$.*

Proof. Filtered colimits are exact, and by **FHHF**, exact functors commute with homology. Hence, filtered colimits commute with homology. $\square$

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